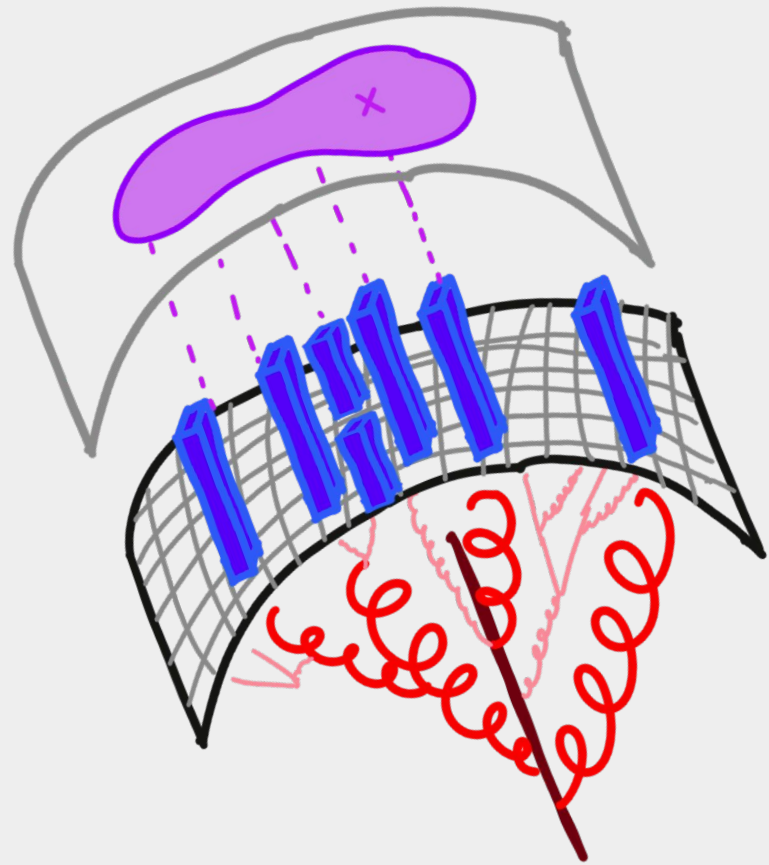


The Shape of Jets

Rikab Gambhir

gambhirb@ucmail.uc.edu



Fermilab Topic of the Week

Based on:

[Ba, Dogra, **RG**, Tasissa, Thaler; [2302.12266](#)]

[**RG**, Thaler, Wu; WIP]

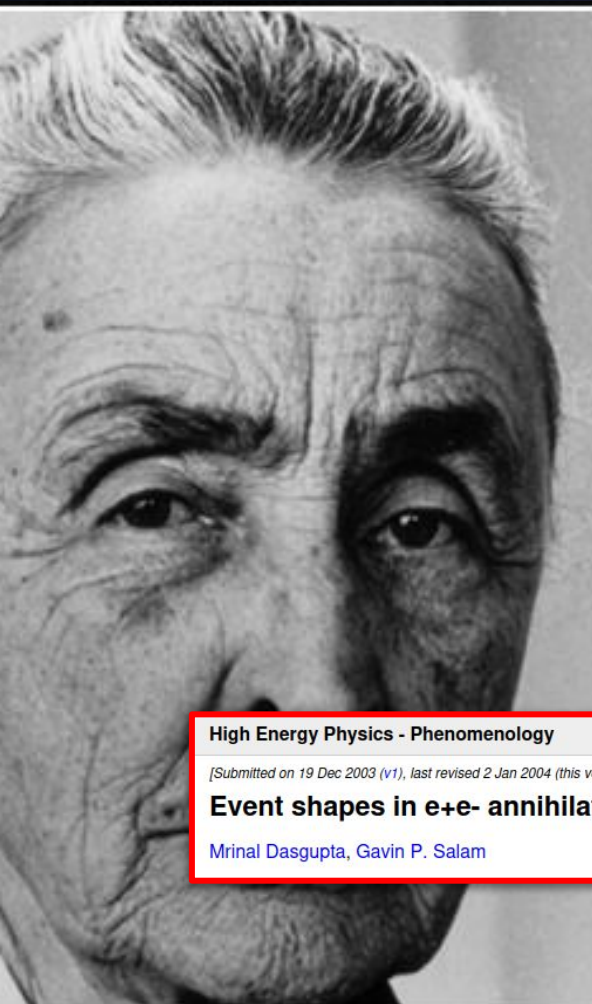
[**RG**, Larkoski, Thaler; [2410.05379](#)]

[**RG**, Osathapan, Thaler; [2403.08854](#)]

[Du Plessis, **RG**; WIP]

[**RG**, LeBlanc, Zhou; [2509.19431](#)]

[**RG**, Larkoski; WIP]



I found I could say things with color
and shapes that I couldn't say any
other way - things I had no words
for.

— Georgia O'Keeffe —

High Energy Physics - Phenomenology

[Submitted on 19 Dec 2003 (v1), last revised 2 Jan 2004 (this version, v2)]

Event shapes in e^+e^- annihilation and deep inelastic scattering

Mrinal Dasgupta, Gavin P. Salam

Number 4

PHYSICS LETTERS

ADVANTAGES OF THE COLOR OCTET GLUON PICTURE[☆]

H. FRITZSCH*, M. GELL-MANN and H. LEUTWYLER**
California Institute of Technology, Pasadena, Calif. 91109, USA

Received 1 October 1973

Fundamental Questions

1. What *can* we measure at colliders for QCD?

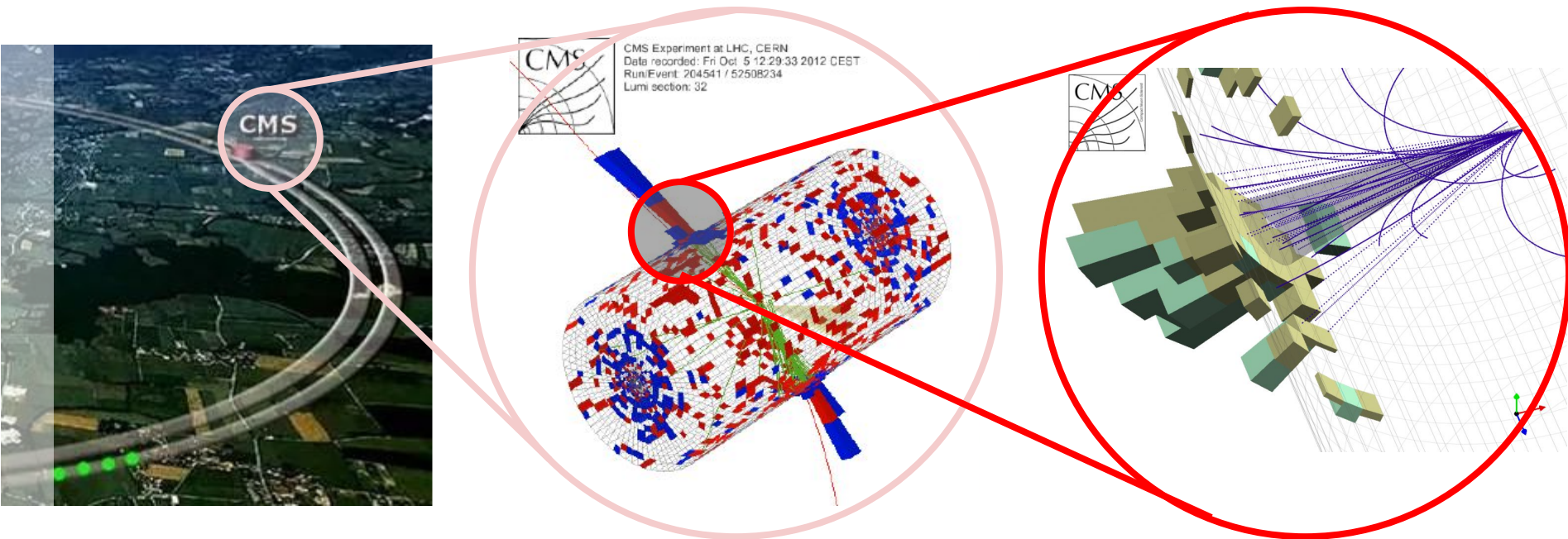
Information Extraction, Energy Flows, Machine Learning...

2. What *should* we measure?

IRC Safety, Robustness, Uncertainties, Interpretability...



Our Setting: **Jets** at the **LHC**



*Not literally the same jet

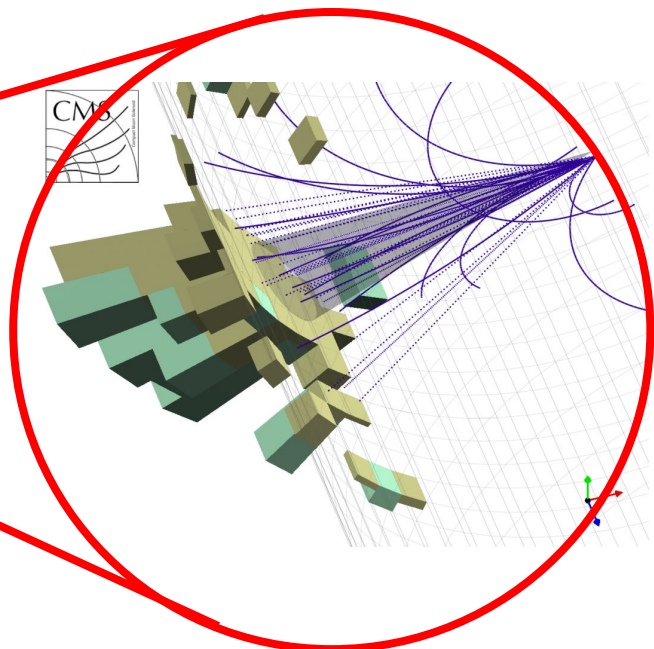
Jets are complicated

We use tools like PYTHIA to predict the distribution of phase space.

$$\frac{d\sigma}{d\Phi} = \text{Distribution of all particles in an event}$$

But with $\mathcal{O}(\frac{Q}{m_\pi})$ 100 to 1000 dimensions, this is a highly exclusive cross section. **Hard to get right!**

- Non perturbative effects
- Large logs, especially when exclusive
- IR divergences
- Detector Effects^{*}



100s-1000s of dimensions
of information *per event*

Exclusive: Hard to predict

^{*}I won't discuss detector effects in this talk much. But please ask me more about using ML to calibrate later!

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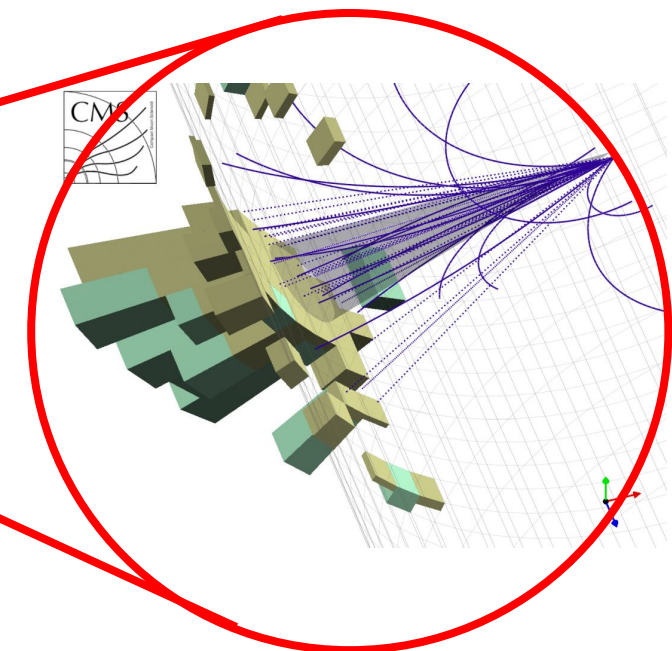
100s-1000s of dimensions
of information *per event*

Exclusive: Hard to predict

Solution: Pick a handful of observables to compress the information. But which ones?

^{*}I won't discuss detector effects in this talk much. But please ask me more about using ML to calibrate later!

~~Jets are complicated~~ can be simple



100s-1000s of dimensions
of information *per event*

Exclusive: Hard to predict



Solution: Compress to observables.

Examples:

- Event Shapes / Substructure
e.g. Thrust, Optimal Transport observables
- ML Observables
e.g. Classifiers, Likelihoods,
- EECs
e.g. See previous talks!
- Multiplicities, Flavors
Not IRC-safe, but important, especially for heavy ion physics?

Question: What functions of phase space do we choose?

(More) Inclusive: Easier to predict

~~Jets are complicated~~ can be simple

Solution: Compress to observables.

Examples:

I'll focus on these for the rest of this talk

- Event Shapes / Substructure
e.g. Thrust, Optimal Transport observables
- ML Observables
e.g. Classifiers, Likelihoods,
- EECs
e.g. See previous talks!
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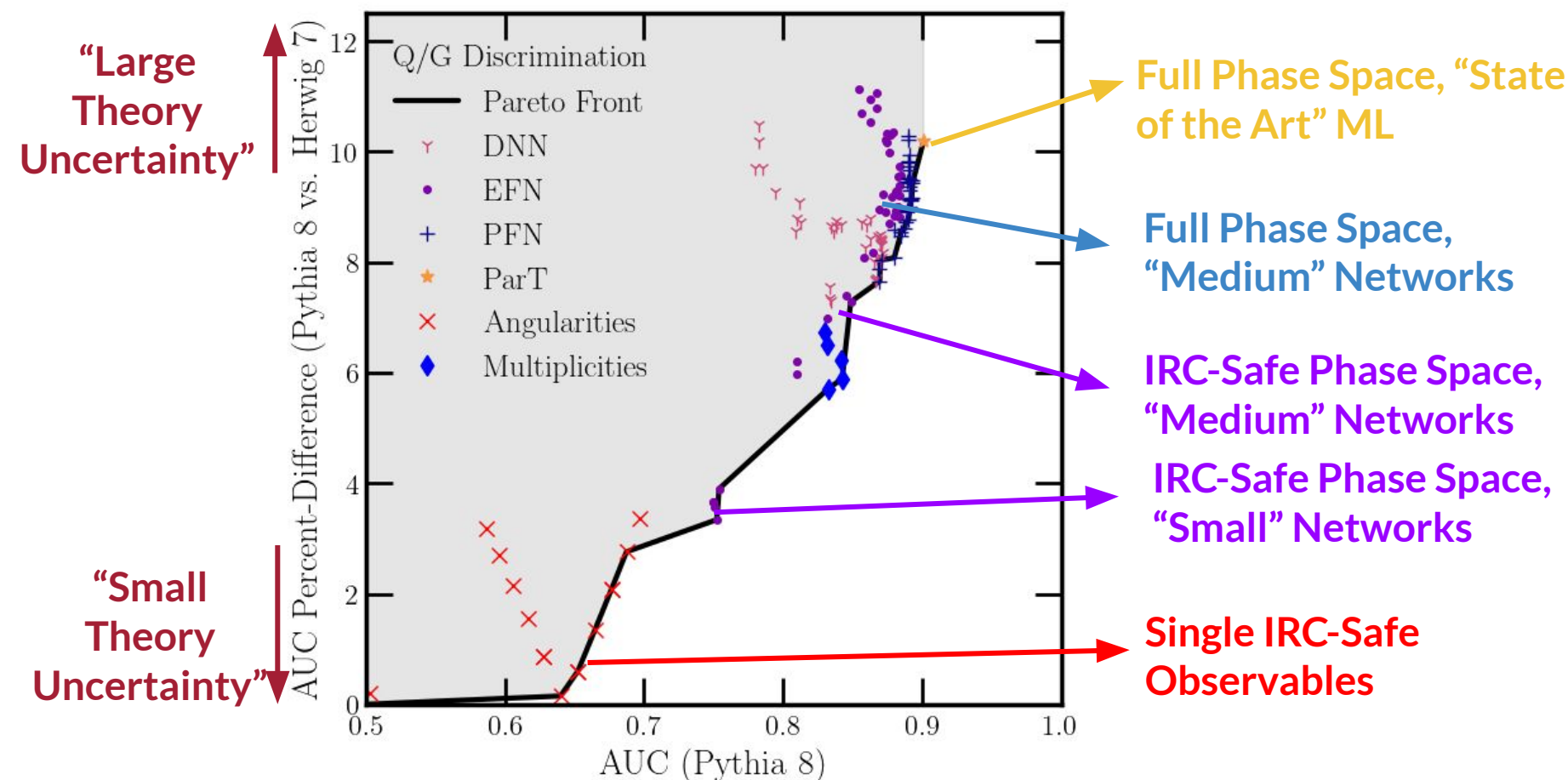
Exclusive: Hard to predict

Question: What functions of phase space do we choose?

(More) Inclusive: Easier to predict

Why This Matters? Example: Jet Tagging

Jet tagging* is bread-and-butter in modern collider analyses, e.g. for isolating pure samples



**Insofar as jet categories are well defined.

***Theory Uncertainty” defined loosely.

Part 0

Intro: **IRC-Safety, Jets, and Geometry**

~~THE UNREASONABLE EFFECTIVENESS
OF MATHEMATICS IN THE NATURAL
SCIENCES~~

Eugene Wigner

The Wasserstein Metric

Collider Physics

Mathematics, rightly viewed, possesses not only truth, but supreme beauty cold and austere, like that of sculpture. It appeals to any part of our weaker nature, without the gorgeous trappings of painting or music, yet sublimely pure, and capable of a stern perfection such as only the greatest art can show. The true spirit of delight, the exaltation, the sense of being more than Man, which is the touchstone of the highest excellence, is to be found in mathematics as surely as in poetry.

- BERTRAND RUSSELL, Study of Mathematics

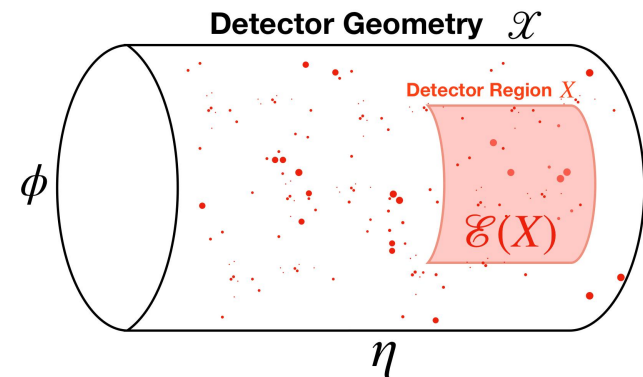
The Energy Flow

What is IRC-safe? Embed event data into the **energy flow**:

$$\mathcal{E}(y) = \sum_i z_i \delta(y - y_i)$$

Detector Coordinate (η, ϕ)

Energy Fraction E_i / E_{Tot}



The **energy flow** contains *ALL* IRC-safe information about an event!

- Claim 1: The energy flow is a continuous embedding of events.
- Claim 2: An observable $\mathcal{O}(\mathcal{E})$ is IRC-safe if and only if it is continuous with respect to the *weak* topology* on energy flows.
- Claim 3: The Wasserstein metric is the only *faithful* distance-preserving metric on energy flows.

*What happened to charge, flavor, mass, and other particle information? Ask me!

**In this talk, will treat “jets” and “events” as the same thing – the difference here is unimportant

The Energy Flow

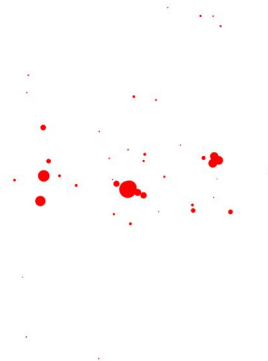
What is IRC-safe? Embed event data into the **energy flow**.

Emotionally, the IRC-safe part of a jet is just a **weighted point cloud**.

And weighted point clouds are just **probability distributions**.

So, asking IRC-safe questions about jets corresponds to asking questions about probability distributions!

What is the *mean* of this jet?
What is [*any moment*] of this jet?
How similar is this jet to [*given distribution*]?
How can I parametrize/fit this distribution?



A PYTHIA-simulated top jet as a point cloud

Claim 1: The energy flow is a continuous embedding of events.

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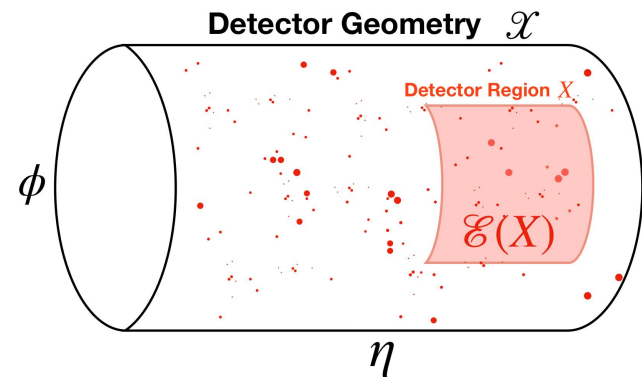
**In this talk, will treat “jets” and “events” as the same thing – the difference here is unimportant

Mathematical Details

Formally, an energy flow is a **measure** on the detector boundary

$$\mathcal{E}(X) = \int_X dx \mathcal{E}(x)$$

“How much total energy did I see in the detector region X ?”



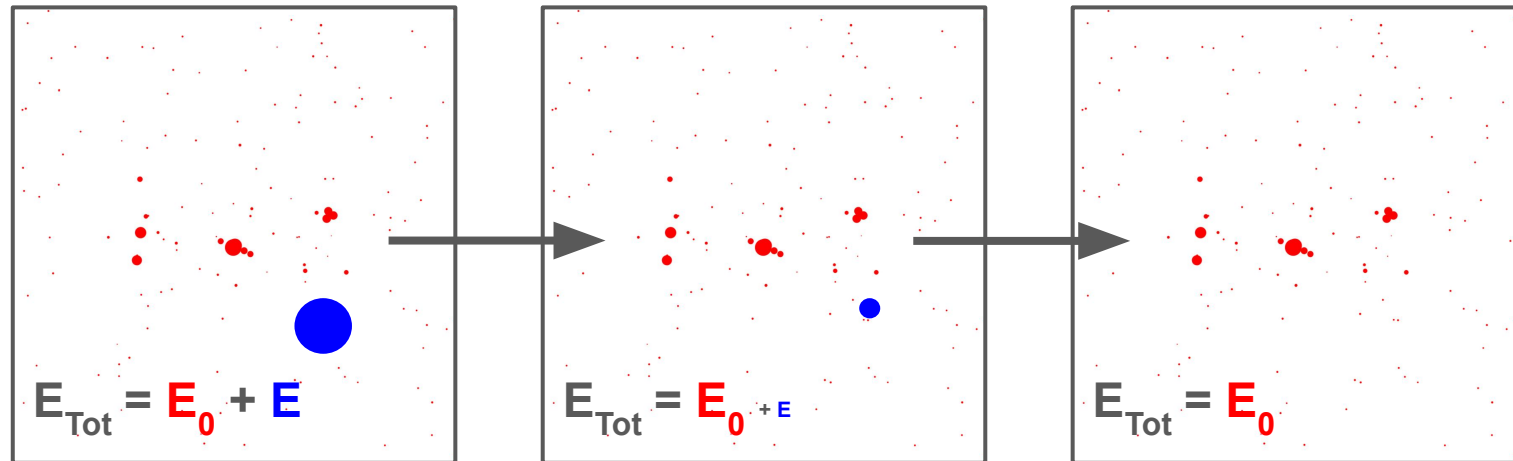
The fundamental operation on measures is **integration**:

$$\langle \mathcal{E}, \phi \rangle \equiv \int_{\mathcal{X}} dx \mathcal{E}(x) \phi(x)$$

“What is the energy-weighted expectation value of ϕ ?”

Mathematical Details – Topology

Definition (The **Weak* Topology**): A sequence of measures converges if all of their expectation values converge, as real numbers.

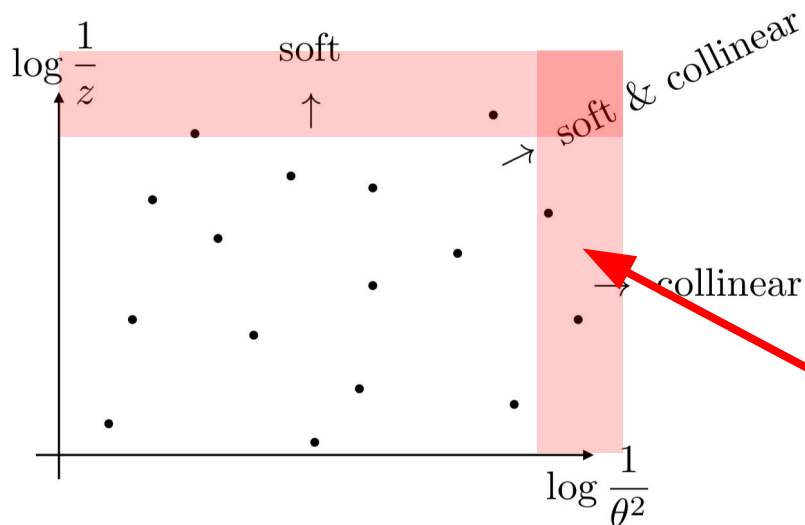


Definition (**Weak* Continuity**): An observable $\mathcal{O}(\mathcal{E})$ is continuous with respect to energy flows if, for any sequence of measures $\mathcal{E}_n$ that converges to $\mathcal{E}$, the sequence of real numbers $\mathcal{O}(\mathcal{E}_n)$ converges to $\mathcal{O}(\mathcal{E})$.

Topology $\Leftrightarrow$ IRC-Safety

Two ways to change the expectation values of an energy flow:

1. Change a particle's energy slightly, or add a low-energy particle - **IR**
2. Move a particle's position slightly, or split particles in two - **C**



An observable $\mathcal{O}$ is continuous if it changes only slightly under the above perturbations.

The regions of phase space causing IRC divergences is suppressed — $\mathcal{O}$ is **IRC-Safe**!

Geometry and the Wasserstein Metric

When are two events (distributions) similar? We need a metric to compare!

Properties we want:

1. ... is non-negative, non-degenerate^{*}, symmetric and finite — is an *actual* metric
 2. ... is weak^{*} continuous (IRC-safe)
 3. ... lifts the detector metric **faithfully** — preserves geometry
- ↙ We will break this soon!
- ↗ Explained shortly!

Geometry and the Wasserstein Metric

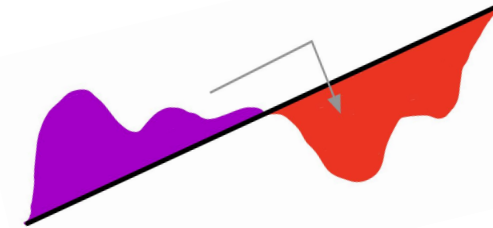
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↑ Explained shortly!

↓ We will break this soon!



EMD = Work done to move "dirt" optimally

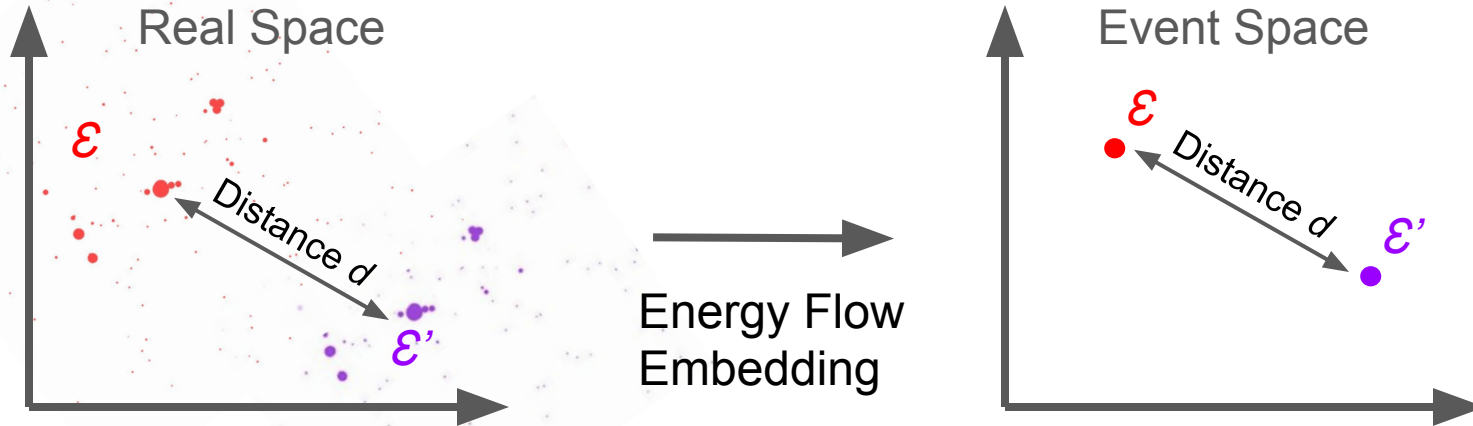
The only^{*} metric on distributions satisfying the above is the **Wasserstein Metric**:

$$\text{EMD}^{(\beta, R)}(\mathcal{E}, \mathcal{E}') = \min_{\pi \in \mathcal{M}(\mathcal{X} \times \mathcal{X})} \left[\frac{1}{\beta R^\beta} \left\langle \pi, d(x, y)^\beta \right\rangle \right] + |\Delta E_{\text{tot}}|$$

$$\pi(\mathcal{X}, Y) \leq \mathcal{E}'(Y), \quad \pi(X, \mathcal{X}) \leq \mathcal{E}(X), \quad \pi(\mathcal{X}, \mathcal{X}) = \min(E_{\text{tot}}, E'_{\text{tot}})$$

*There exist other metrics on distributions that are faithful only for very specific real-space distance norms, but we want them all!

The Importance of Being **Faithful***



A metric on events is **faithful** if, whenever two otherwise identical events $\mathcal{E}$ and $\mathcal{E}'$ are separated in real space by a distance d , the distance between the events is also d . Or d to a constant power

Only the **Wasserstein Metric** does this! → Can use to build **event** and **jet shapes** (old and new)!

Faithfulness also ensures very nice **numerical properties**, including no vanishing or exploding gradients.

Are there **other Metrics?** Yes.

If we:

1. Only require *almost always a metric* — that is, only require $d(x,y) = 0$ *almost always* implies $x = y$.
2. Replace *faithfulness* with *isometry preserving* — that is $d(x,y) = 0$ if x and y are related by a translation, rotation, or mirror

Then we can write another Wasserstein-Based metric, called the **Spectral EMD**

$$\begin{aligned} \text{SEMD}_{\beta,p=2}(s_A, s_B) = & \sum_{i < j \in \mathcal{E}_A} 2E_i E_j \omega_{ij}^2 + \sum_{i < j \in \mathcal{E}_B} 2E_i E_j \omega_{ij}^2 \\ & - 2 \sum_{n \in \mathcal{E}_A^2, l \in \mathcal{E}_B^2} \omega_n \omega_l \left(\min [S_A(\omega_n^+), S_B(\omega_l^+)] - \max [S_A(\omega_n^-), S_B(\omega_l^-)] \right) \\ & \times \Theta (S_A(\omega_n^+) - S_B(\omega_l^-)) \Theta (S_B(\omega_l^+) - S_A(\omega_n^-)) , \end{aligned}$$

Unlike the EMD, the SEMD is very easy to *evaluate exactly and quickly!*

Are there **other Metrics?** Yes. Even More!

[[2405.16698](#)]

Phase Space Metrics

Not IRC-Safe

$$d\Pi_N = \frac{(2\pi)^4}{Q^4} \left(\frac{Q^2}{16\pi^3} \right)^N d\Delta_{N-1} dS^{2N-3}$$

Factorizes phase space
into (simplex) x
(hypersphere)

[[2111.03670](#)]

Unbalanced OT

$$\begin{aligned} & \text{HK}^\kappa(\mathcal{E}, \mathcal{E}') \\ &= \min_{\gamma_{ij} \geq 0} \sum_{ij} (\ell^\kappa(d_{ij})\gamma_{ij} + \kappa^2 \text{KL}(\mathcal{G}, \mathcal{E}) + \kappa^2 \text{KL}(\mathcal{G}', \mathcal{E}'))^{1/2} \end{aligned} \quad (8)$$

Hellinger distance

Different variations of OT
when events are not
normalized

[[2008.08604](#)]

Linearized OT

$$\text{LOT}_{r, \tilde{r}}(\mathcal{E}, \tilde{\mathcal{E}}) = \left(\sum_i R_i \|z_i - \tilde{z}_i\|^2 \right)^{1/2}.$$

$$\lim_{N \rightarrow +\infty} \text{LOT}_{r^N, \tilde{r}^N}(\mathcal{E}, \tilde{\mathcal{E}}) = W_{2, \mathcal{R}}(\mathcal{E}, \tilde{\mathcal{E}}).$$

Fast, Riemannian
approximations to
Wasserstein

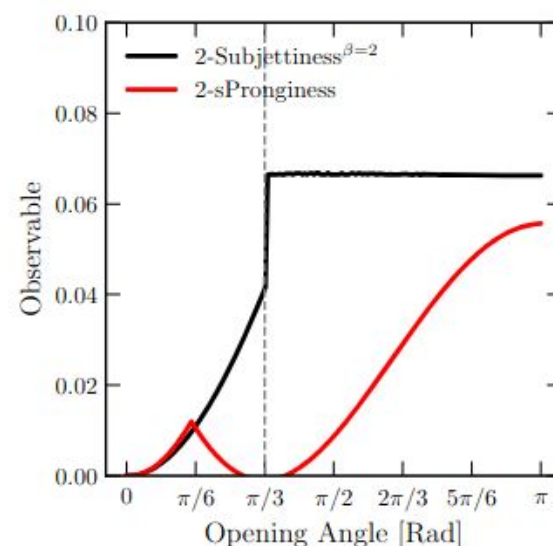
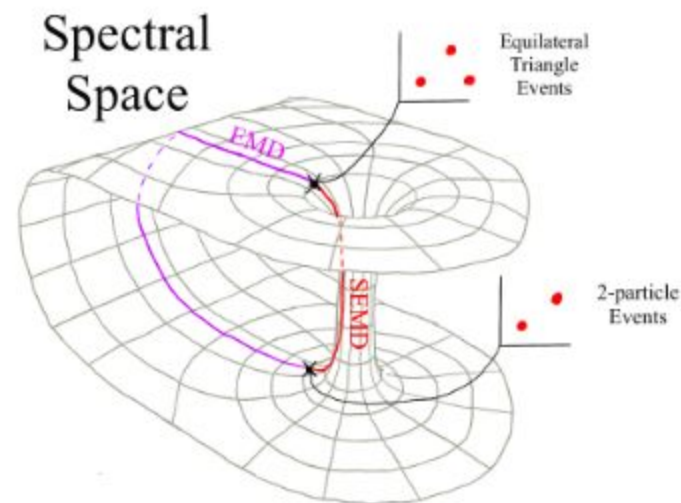
The geometry of events is an active area of research!
Ask me afterwards about “**Theory Space OT**”...

SEMD and Topology

The SEMD and EMD agree *locally*, but because the SEMD has measure-0 degeneracies, they can differ wildly for some events.

Two events far apart in EMD space can be extremely close in SEMD space, which changes the topology of the space.

However, the IRC structure is the same.



Part I

Can you Hear the **Shape** of a Jet?



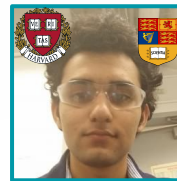
Download
SHAPER!

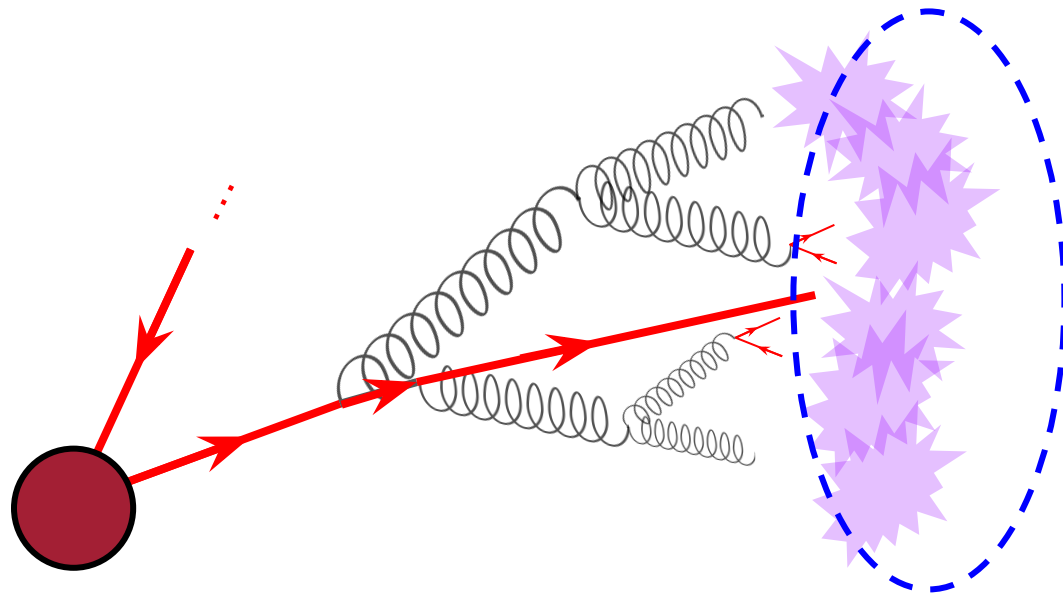


Download
SPECTER!

Try `pip install pyshaper`

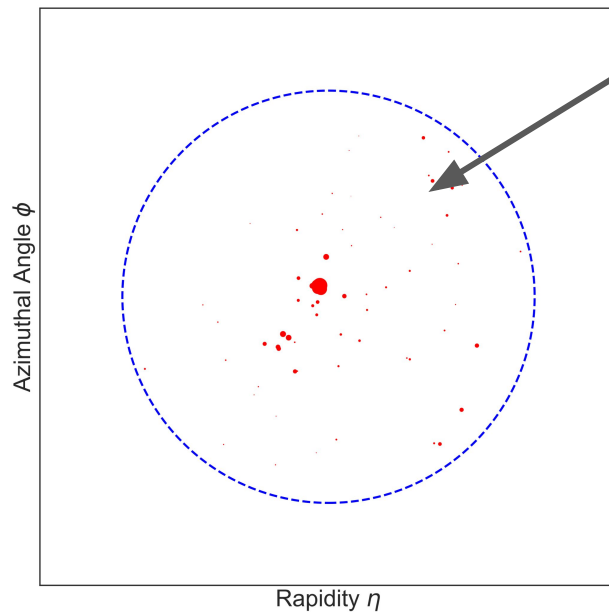
(New!) `pip install specterpy`





How “*big*” is this jet?

... What do we even mean by the “size” of a jet?



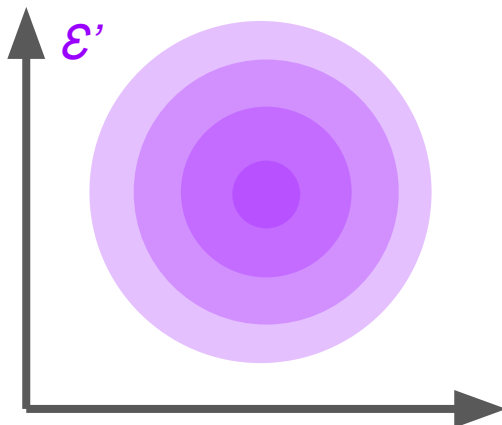
By following this line of thinking, we'll be able to come up with a new class of QCD observables!

I don't mean the fixed size of a jet algorithm, e.g. AK4, but rather a dynamical notion of a jet size!

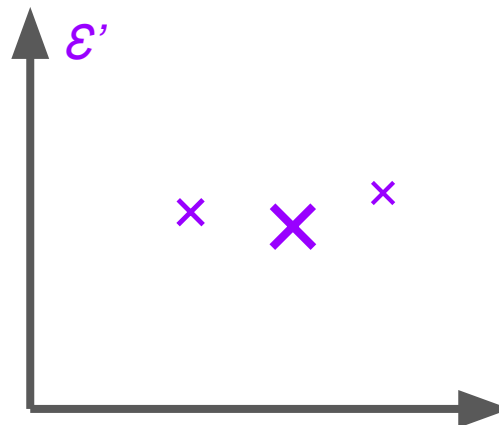
Shapes as Energy Flows

Energy flows don't have to be real events – they can be *any* energy distribution in detector space, or **shape**. In our framework, these shapes will later become our **Function Space**

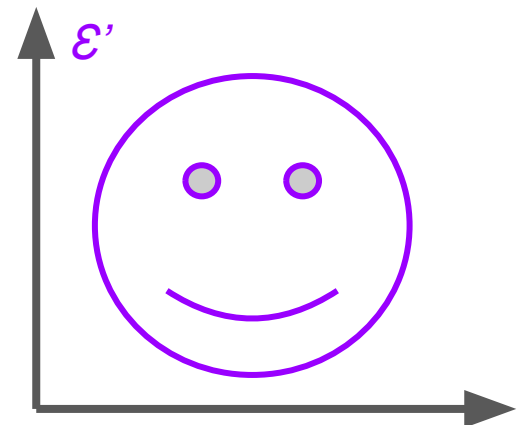
Can make anything you want! Even continuous or complicated shapes.
(Or, something you calculate in perturbative QCD)



Shape = 2D Gaussian



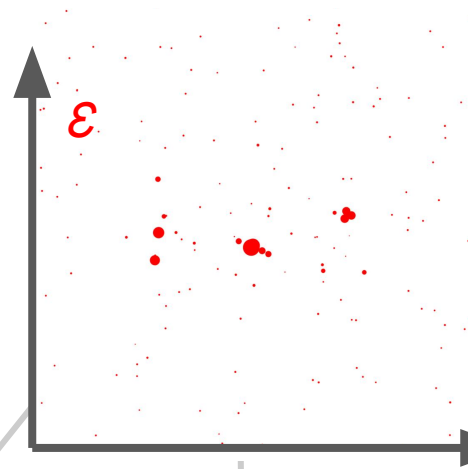
Shape = 3 Points



Shape = Smile

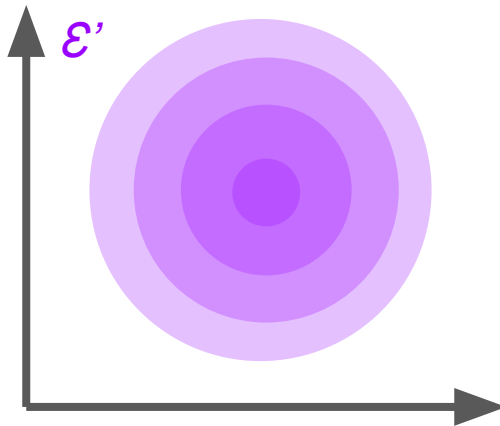
Shapiness

The EMD between a real event or jet $\mathcal{E}$ and idealized shape $\mathcal{E}'$ is the [shape]iness of $\mathcal{E}$ – a well defined IRC-safe observable!



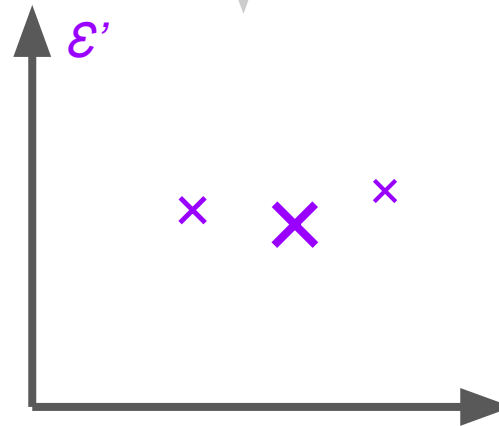
Answers the question:
“How much like the shape $\mathcal{E}'$ is my event $\mathcal{E}$?”

Med EMD($\mathcal{E}, \mathcal{E}'$)
“Gaussiness”



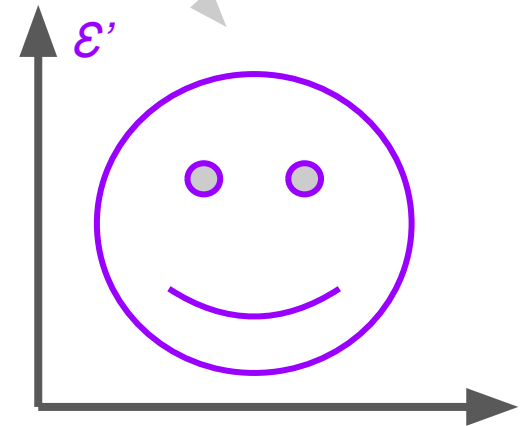
Shape = 2D Gaussian

Low EMD($\mathcal{E}, \mathcal{E}'$)
“3-Pointiness”
AKA “3-Subjettiness”



Shape = 3 Points

High EMD($\mathcal{E}, \mathcal{E}'$)
“Smileyiness”



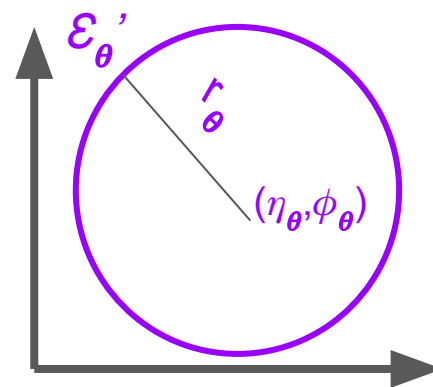
Shape = Smile

Mathematical Details - Shapiness

Rather than a single shape, consider a **parameterized manifold** $\mathcal{M}$ of **energy flows**.

e.g. The manifold of uniform circle energy flows:

$$\mathcal{E}_\theta'(y) = \begin{cases} \frac{1}{2\pi r_\theta} & |\vec{y} - \vec{y}_\theta| = r_\theta \\ 0 & |\vec{y} - \vec{y}_\theta| \neq r_\theta \end{cases}$$



Then, for an event $\mathcal{E}$, define the **shapiness** $\mathcal{O}_{\mathcal{M}}$ and **shape parameters** $\theta_{\mathcal{M}}$, given by:

$$\mathcal{O}_{\mathcal{M}}(\mathcal{E}) \equiv \min_{\mathcal{E}_\theta \in \mathcal{M}} \text{EMD}^{(\beta, R)}(\mathcal{E}, \mathcal{E}_\theta)$$

$$\theta_{\mathcal{M}}(\mathcal{E}) \equiv \operatorname{argmin}_{\mathcal{E}_\theta \in \mathcal{M}} \text{EMD}^{(\beta, R)}(\mathcal{E}, \mathcal{E}_\theta)$$

Similar to a GAN, but we have only a few parameters

Observables $\Leftrightarrow$ Manifolds of Shapes

Observables can be specified solely by defining $\mathcal{O}_{\mathcal{M}}(\mathcal{E}) \equiv \min_{\mathcal{E}_{\theta} \in \mathcal{M}} \text{EMD}^{(\beta, R)}(\mathcal{E}, \mathcal{E}_{\theta})$,

a **manifold of shapes**:

$$\theta_{\mathcal{M}}(\mathcal{E}) \equiv \operatorname{argmin}_{\mathcal{E}_{\theta} \in \mathcal{M}} \text{EMD}^{(\beta, R)}(\mathcal{E}, \mathcal{E}_{\theta}),$$

Functional Minimization! Many well-known observables* already have this form!

Observable	Manifold of Shapes
N -Subjettiness	Manifold of N -point events
N -Jettiness	Manifold of N -point events with floating total energy
Thrust	Manifold of back-to-back point events
Event / Jet Isotropy	Manifold of the single uniform event

... and more!

All of the form “How much like **[shape]** does my **event** look like?”

Generalize to *any* shape.

*These observables are usually called event shapes or jet shapes in the literature – we are making this literal!

Hearing Jets Ring

(and Disk, and Ellipse)

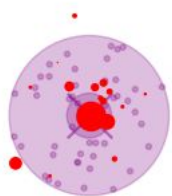
$\mathcal{O}_{\mathcal{M}}(\mathcal{E})$ answers: “How much like a **shape** in $\mathcal{M}$ does my **event** $\mathcal{E}$ look like?”

$\theta_{\mathcal{M}}(\mathcal{E})$ answers: “Which **shape** in $\mathcal{M}$ does my **event** $\mathcal{E}$ look like?”

Can define complex manifolds to probe increasingly subtle geometric structure!

Shape	Specification	Illustration
Ringiness $\mathcal{O}_R$	Manifold of Rings $\mathcal{E}_{x_0, R_0}(x) = \frac{1}{2\pi R_0}$ for $ x - x_0 = R_0$ $x_0 = \text{Center}, R_0 = \text{Radius}$	
Diskiness $\mathcal{O}_D$	Manifold of Disks $\mathcal{E}_{x_0, R_0}(x) = \frac{1}{\pi R_0^2}$ for $ x - x_0 \leq R_0$ $x_0 = \text{Center}, R_0 = \text{Radius}$	
Ellipsiness $\mathcal{O}_E$	Manifold of Ellipses $\mathcal{E}_{x_0, a, b, \varphi}(x) = \frac{1}{\pi ab}$ for $x \in \text{Ellipse}_{x_0, a, b, \varphi}$ $x_0 = \text{Center}, a, b = \text{Semi-axes}, \varphi = \text{Tilt}$	
(Ellipse Plus Point)iness	Composite Shape $\mathcal{O}_E \oplus \tau_1$ Fixed to same center x_0	
N-(Ellipse Plus Point)iness Plus Pileup	Composite Shape $N \times (\mathcal{O}_E \oplus \tau_1) \oplus \mathcal{I}$	

Some examples of new shapes you can define!



Disk + δ -function



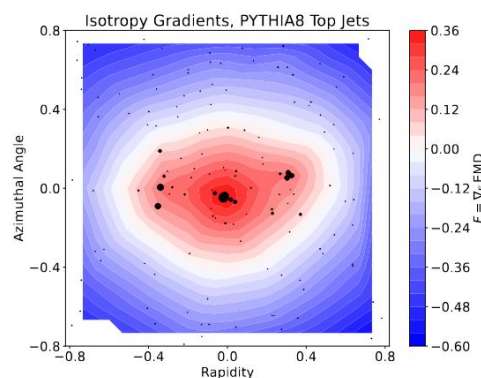
Ring + δ -function

Probing **collinear** (δ -function) and **soft** (shape) structure!

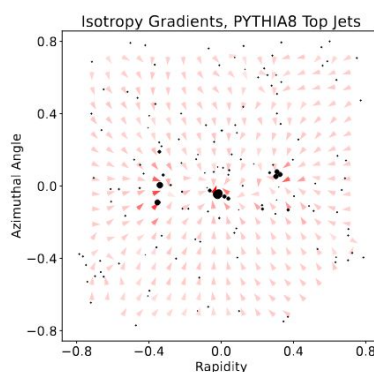
Aside: Approximating the EMD

The Wasserstein Metric is notoriously hard to evaluate, and harder to optimize over. Instead we use the **Sinkhorn Divergence** which approaches the Wasserstein Metric as $\epsilon \rightarrow 0$.

$$\text{OT}_{\epsilon}^{(\beta, R)}(\mathcal{E}, \mathcal{E}') = \max_{f, g: \mathcal{X} \rightarrow \mathbb{R}} \left[\langle \mathcal{E}, f \rangle + \langle \mathcal{E}', g \rangle - \epsilon^{\beta} \log \left\langle \mathcal{E}(x) \otimes \mathcal{E}'(y), e^{\left(\frac{1}{\epsilon^{\beta}} (f(x) + g(y)) - \frac{1}{\beta R^{\beta}} d(x, y)^{\beta} \right)} \right\rangle \right]$$
$$S_{\epsilon}(\mathcal{E}, \mathcal{E}') = \text{OT}_{\epsilon}(\mathcal{E}, \mathcal{E}') - \frac{1}{2} \text{OT}_{\epsilon}(\mathcal{E}, \mathcal{E}) - \frac{1}{2} \text{OT}_{\epsilon}(\mathcal{E}', \mathcal{E}') \quad \text{“Debiased-Sinkhorn”} [\text{1810.08278}]$$



(a)



(b)

Allows for easy computation of gradients with respect to positions and energies — **we can compute the gradient of a number with respect to an entire jet!**

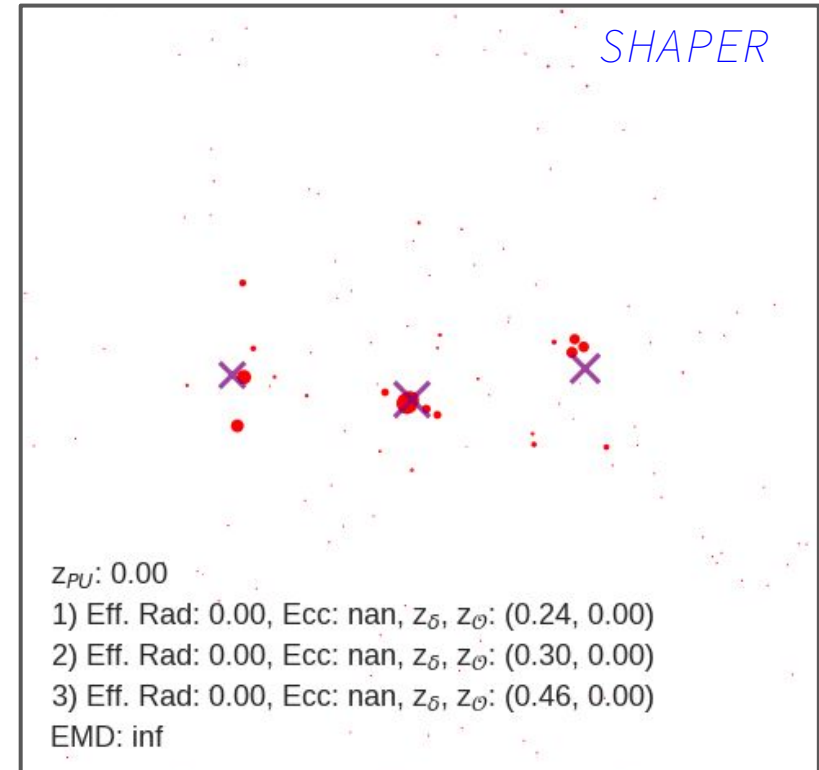
New IRC-Safe Observables

The *SHAPER* framework makes it easy to algorithmically invent new jet observables!

e.g. *N-(Ellipse+Point)iness+Pileup* as a jet algorithm:

- Learn jet centers + collinear radiation
- Dynamic jet radii (no R parameter!)
- Dynamic eccentricities and angles
- Dynamic jet energies
- Dynamic pileup (no z_{cut} parameter!!!)
- Learned parameters for discrimination

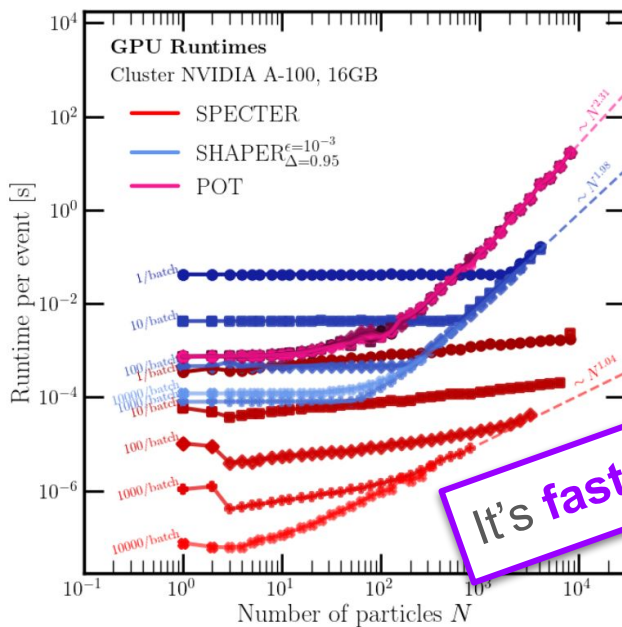
Can design custom specialized jet algorithms to learn jet substructure!



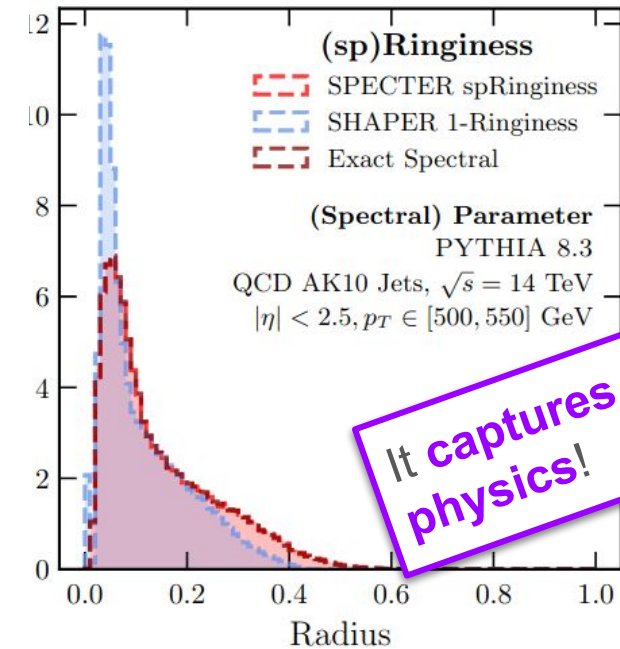
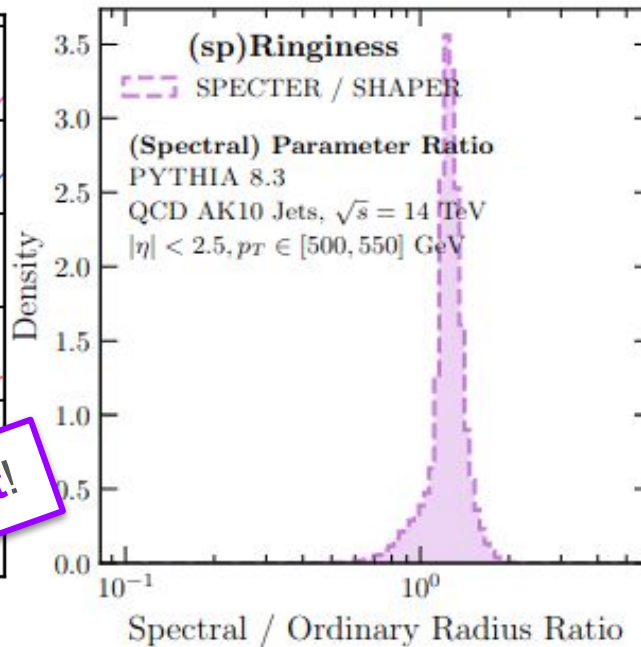


SPECTER: OT, but *FAST*

By demanding our loss is only *almost** always a metric on events rather than always a metric, we can build a new loss that is *completely and exactly solvable*!



It's fast!



It captures physics!



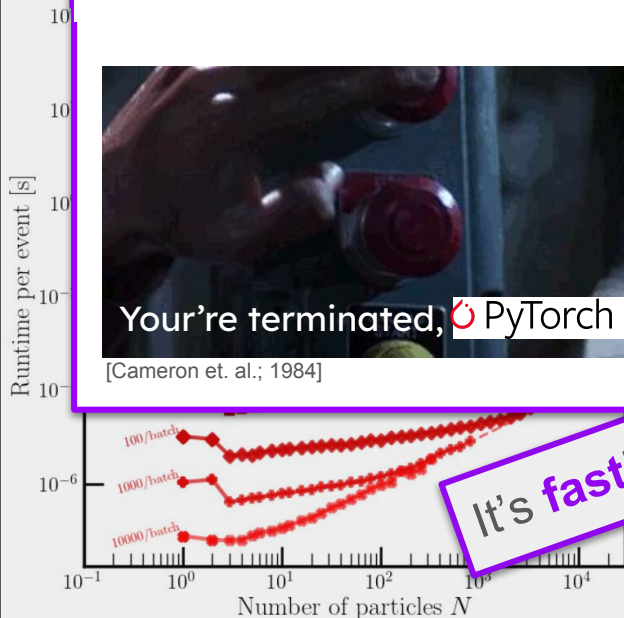
SPECTER: OT, but FAST

Exact and closed form
Spectral Jet Radius!
No gradient descent or
approximations required!

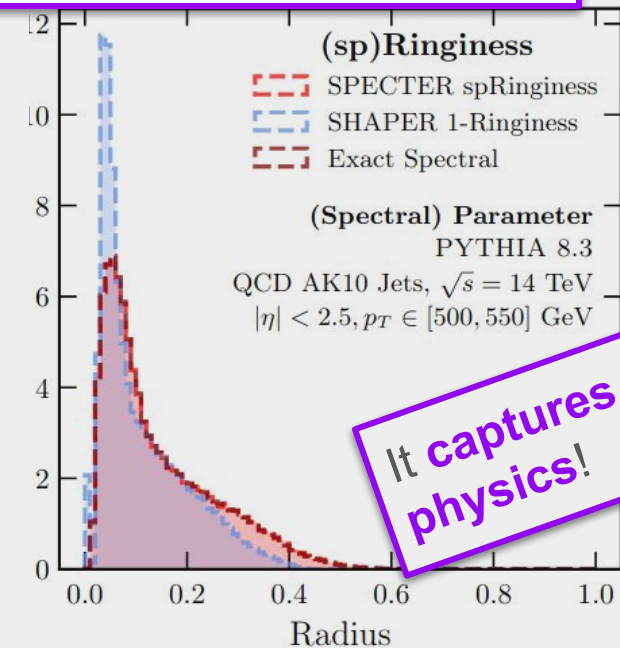
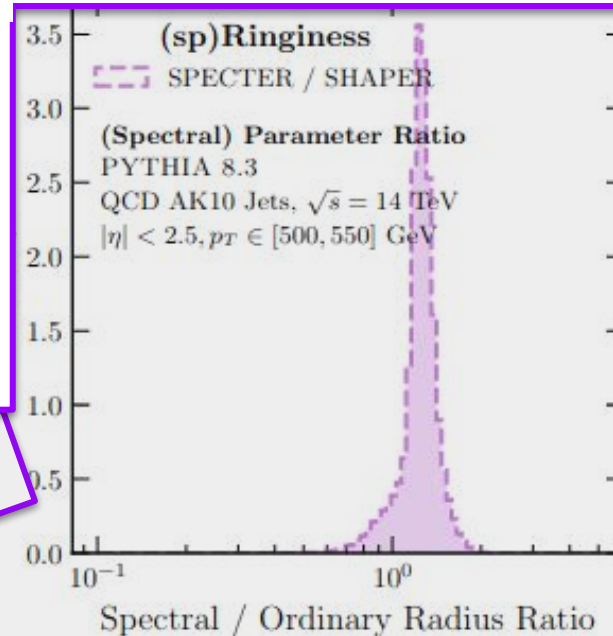
$$r_{\text{opt}} = \frac{2}{\pi} \sum_{\substack{n \in \mathcal{E}^2 \\ \omega_n < \omega_{n+1}}} \omega_n \left[\cos \left(\frac{\pi}{2E_{\text{tot}}^2} S^-(\omega_n) \right) - \cos \left(\frac{\pi}{2E_{\text{tot}}^2} S^+(\omega_n) \right) \right]$$



[Cameron et. al.; 1984]



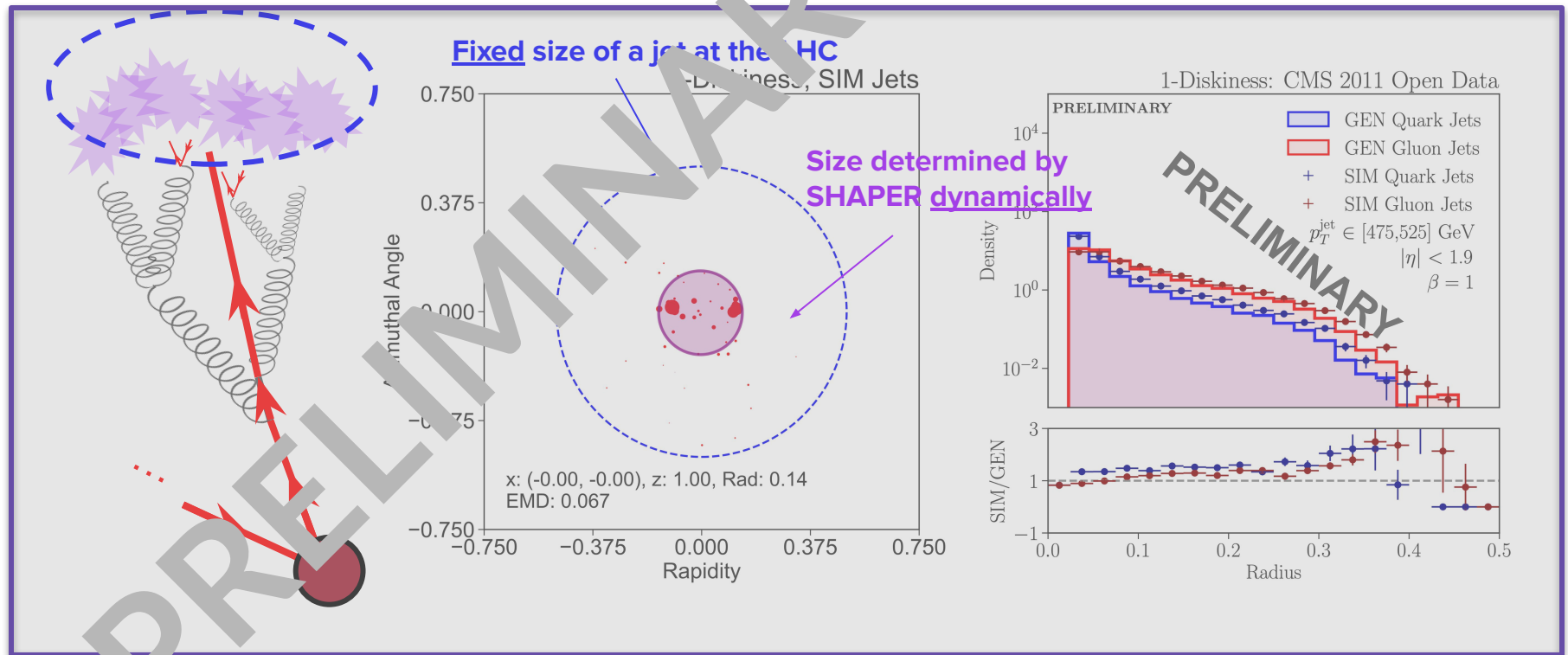
It's fast!



It captures physics!

Back to our question: How Big are Jets?

Slide images from [Wu; APS 2023]



We can now answer this question in a precise way!



What else? Heavy Ion Jets

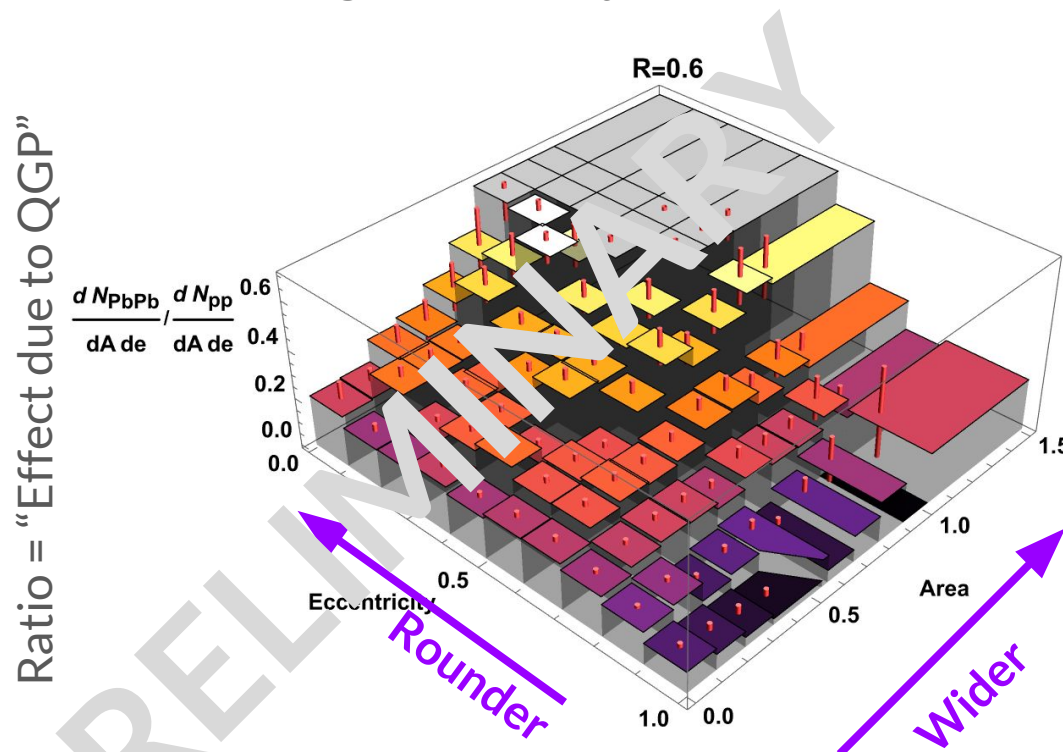
Exploring the shape of jets in Heavy Ion Collisions

The quark-gluon plasma is expected to change the (distribution of the) shape of jets.

The survival rate depends on both the *area* of the jet and the *eccentricity*.

We can directly see that the medium makes jets *wider* and *rounder*.

Fitting **ellipses*** to jets in **Pb-Pb** and **pp**



*These ellipses are not uniform, but rather have exponential falloff from the center

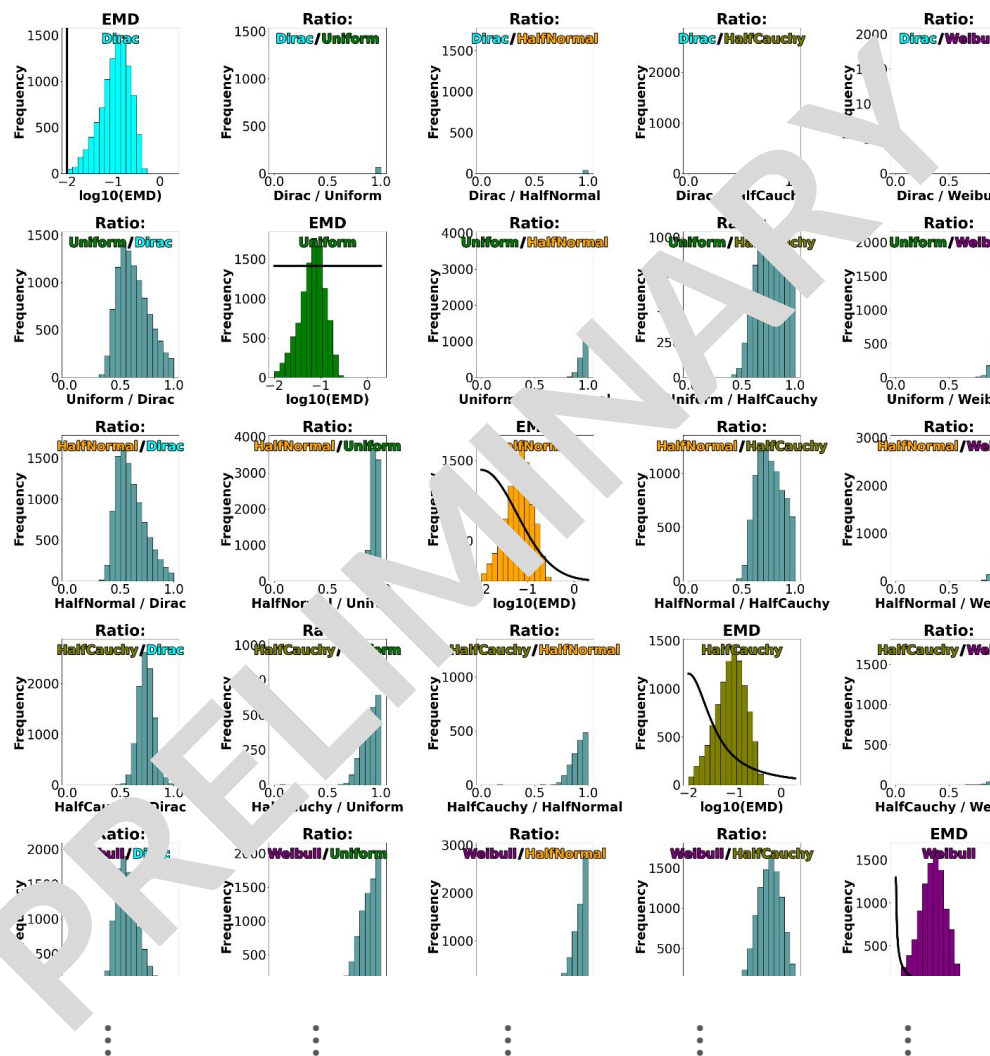
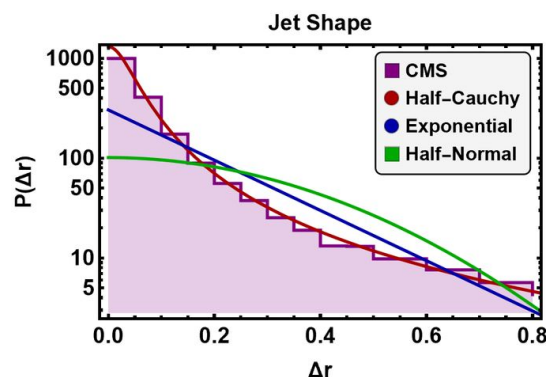
What else? Heavy Ion Jet Profiles

What is the *best* jet shape?

Important distinction: Shape of the *average* jet versus *average* shape of a jet (plotted)

No single simple distribution can capture all jets.

The shape of the *average jet* is close to half-Cauchy, even though any given jet is not!



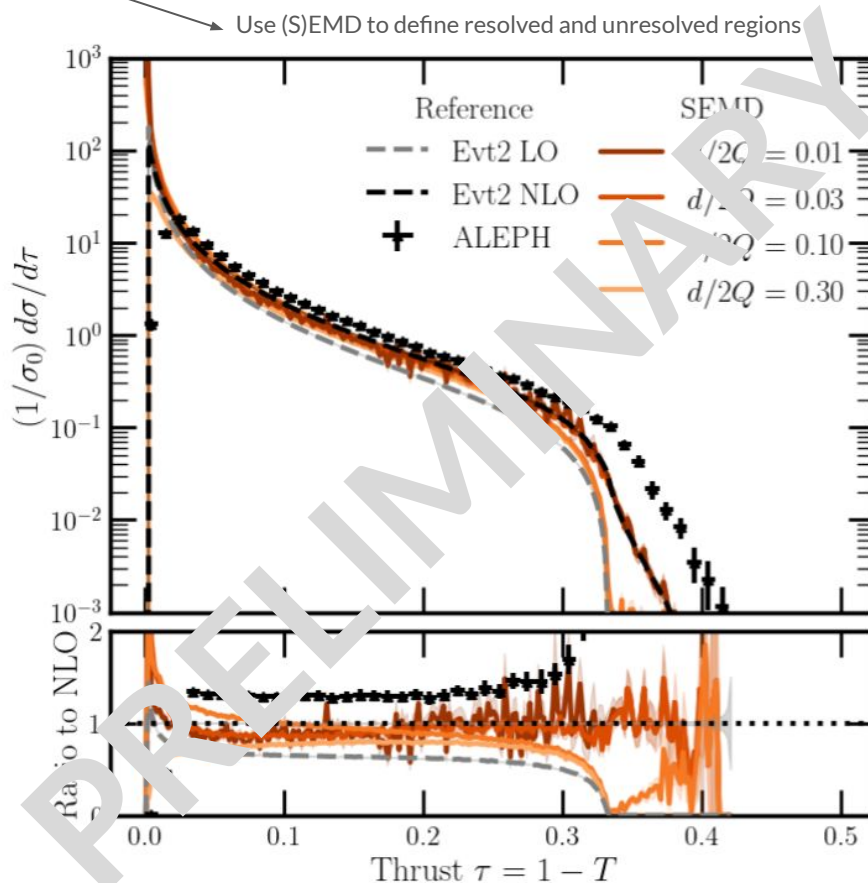
What else? Event Generation

$$\frac{d\sigma^{(\text{NLO})}}{d\mathcal{O}} = \int d\Pi_0 \left\{ |\mathcal{M}^{(0)}(\Pi_0)|^2 + 2 \Re \left(\mathcal{M}^{(0)}(\Pi_0) \mathcal{M}_V^{(1)}(\Pi_0)^* \right) + \text{Sub}(\Pi_0) \right\} \delta \left(\mathcal{O} - \hat{\mathcal{O}}(\Pi_0) \right) \\ + \int d\Pi_1 \left\{ |\mathcal{M}_R^{(1)}(\Pi_1)|^2 - \text{Sub}(\Pi_1) \right\} \delta \left(\mathcal{O} - \hat{\mathcal{O}}(\Pi_1) \right)$$

Perturbative QCD is plagued by **divergences** and **negative weights** — even if everything eventually cancels out, it makes Monte Carlo hard.

Can *regulate* (phase space slice) with EMDs: fast, and being a metric ensures the smallest amount of distorted phase space.

Eat a small power law correction, but fast, positive, finite MC generation!

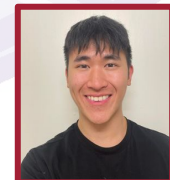


Part II

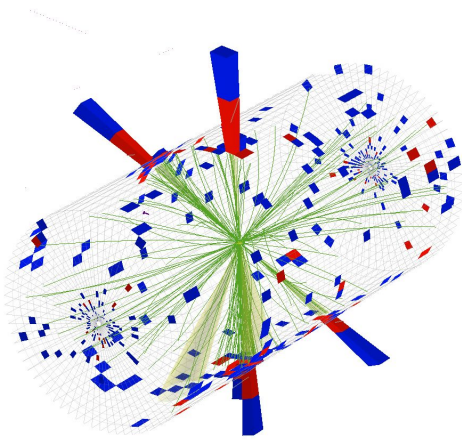
Moments of Clarity: Streamlining Latent Spaces



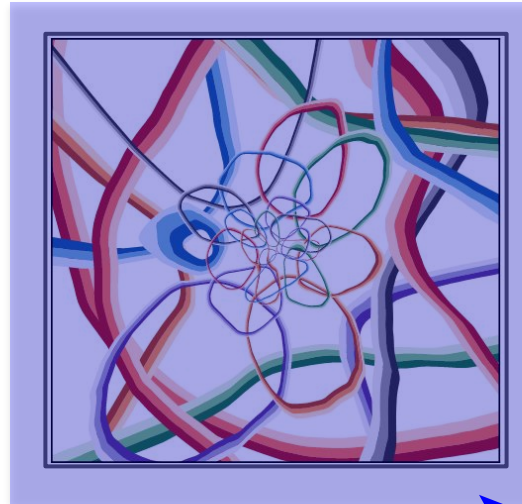
Download
Moment EFNs!



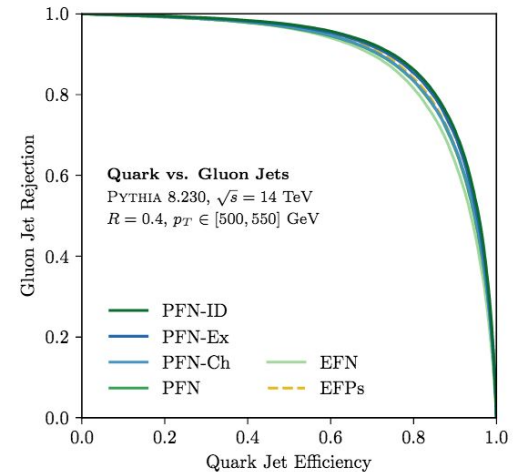
Typical Machine Learning Setup



Collider Data
~1000 Dimensional



Latent Space
~10-100 Dimensional



Observables
~1-10 Dimensional

Pictured: An Energy Flow Network (**EFN**):

$$\mathcal{O}(\{p_1, \dots, p_M\}) = F \left(\sum_{i=1}^M z_i \Phi(\hat{p}_i) \right)$$

(How) can we understand and constrain this?

(How) can we be more efficient?

Deep Learning on Point Clouds

The **Deep Sets Theorem** tells us how to parameterize functions on point clouds*:

Set of momenta $\rightarrow$ $a = 1 \dots L$, the *Latent Dimension* index

$$\mathcal{O}(\mathcal{P}) = F(\langle \phi^a \rangle_{\mathcal{P}})$$

The Deep Sets Theorem guarantees that any function on sets $\mathcal{P}$ can be written this way, for “sufficiently complex” F and ϕ and large enough L .

$\langle \phi \rangle_{\mathcal{P}} \equiv \sum_i z_i \phi(\hat{p}_i)$

For this talk, I will focus primarily on Energy Flow Networks (EFNs) where the sums are energy-weighted.

*More sophisticated architectures still reduce to Deep Sets in some limit, e.g. graph networks

Deep Learning on Point Clouds

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$\langle \phi \rangle_{\mathcal{P}} \equiv \sum_i z_i \phi(\hat{p}_i)$

But how complex do F and ϕ need to be? Can I reduce the number of latent dimensions?

For this talk, I will focus primarily on Energy Flow Networks (EFNs) where the sums are energy-weighted.

*More sophisticated architectures still reduce to Deep Sets in some limit, e.g. graph networks, transformers

Moment Pooling

$$\mathcal{O}(\mathcal{P}) = F(\langle \phi^a \rangle_{\mathcal{P}})$$



Generalize to more moments! “**Moment Pooling**”

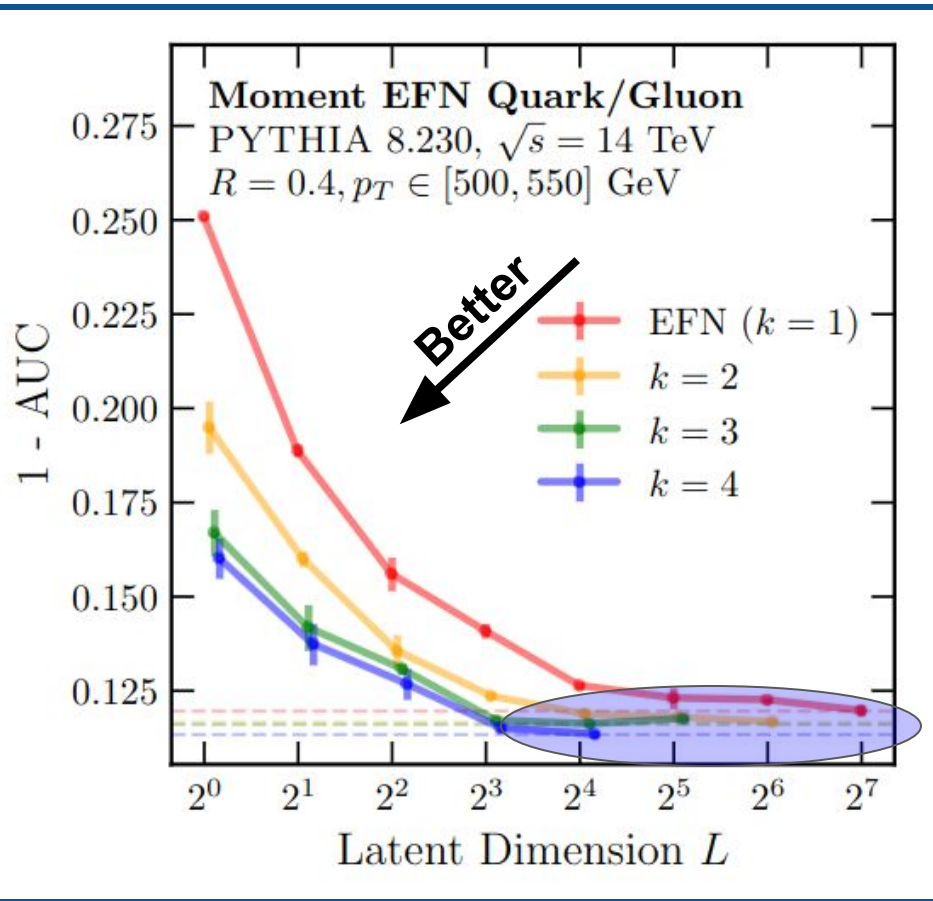
$$\mathcal{O}_k(\mathcal{P}) = F_k(\underbrace{\langle \phi^a \rangle_{\mathcal{P}}}_{1^{\text{st}} \text{ Moment}}, \underbrace{\langle \phi^{a_1} \phi^{a_2} \rangle_{\mathcal{P}}}_{2^{\text{nd}} \text{ Moment}}, \dots, \underbrace{\langle \phi^{a_1} \dots \phi^{a_k} \rangle_{\mathcal{P}}}_{k^{\text{th}} \text{ Moment}})$$

k = Highest order moment considered

$$L_{\text{eff}} = \binom{L+k}{k}$$

Generalized Moments of jets are interesting IRC-safe observables, so let's use them as an ML basis!

Moment Pooling Results

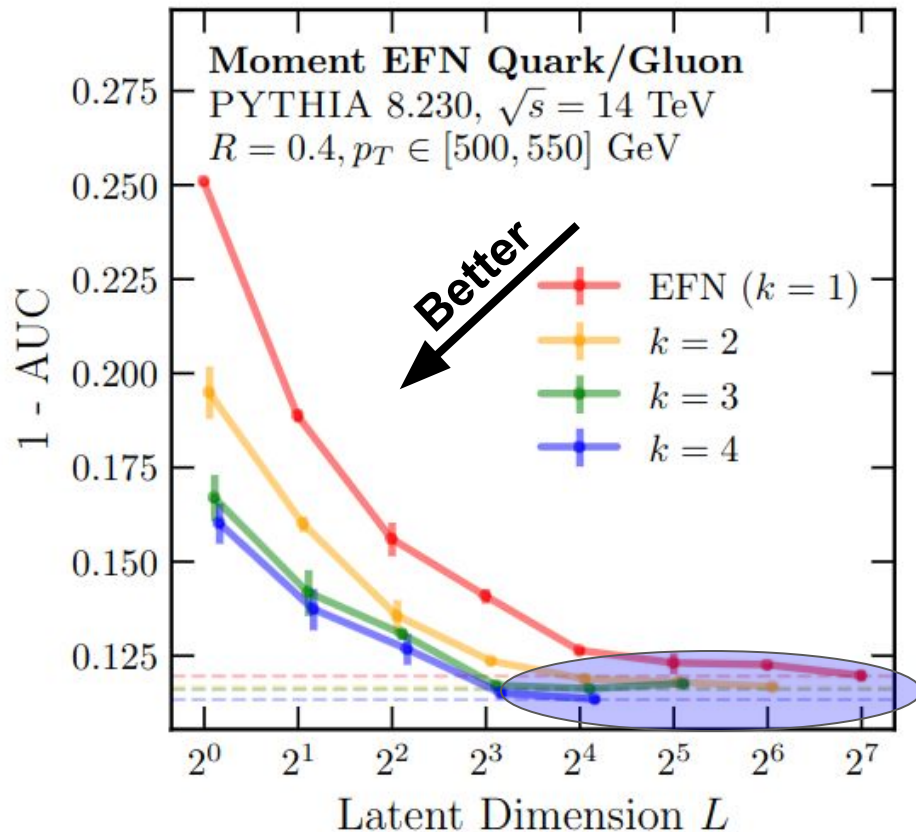


A high order **Moment EFN** can achieve the same top performance on quark/gluon classification as an **ordinary EFN**, but with far fewer latent dimensions!*

A Moment EFN is a more efficient encoding of the same data!

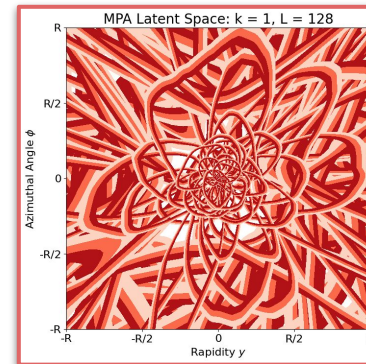
*Interestingly, the performance is constant in the *effective* latent dimensions and number of parameters. Ask me about this!

The Black Box

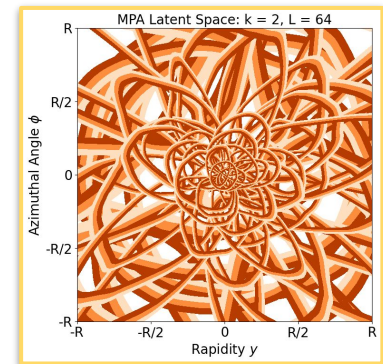


Latent Spaces

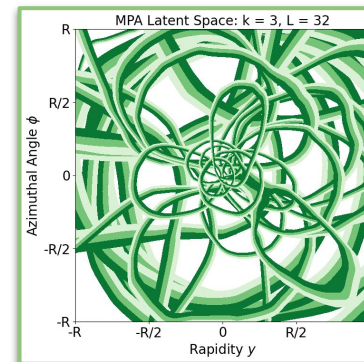
EFN ($k = 1$)
 $L = 128$



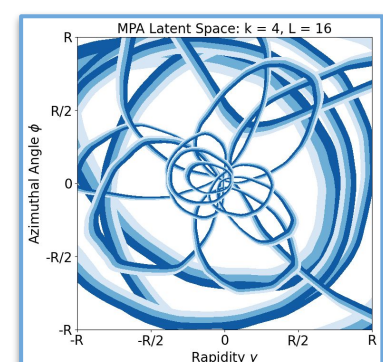
$k = 2$
 $L = 64$



$k = 3$
 $L = 32$



$k = 4$
 $L = 16$

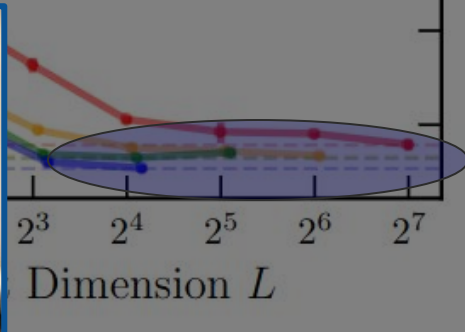


The 45% - 55% contours of each of the L different ϕ functions

*Interestingly, the performance is constant in the *effective* latent dimensions and number of parameters. Ask me about this!

So what? The information is just being shuffled around. Is 16 latent dimensions any more interpretable than 128?

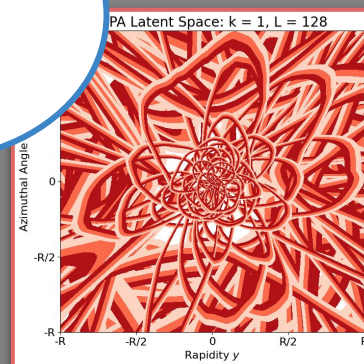
The machine probably learned something, but I didn't!



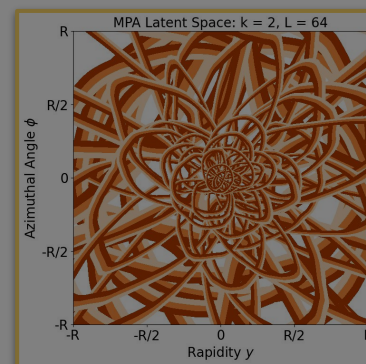
Me, just before the **moment** of clarity.

Latent Spaces

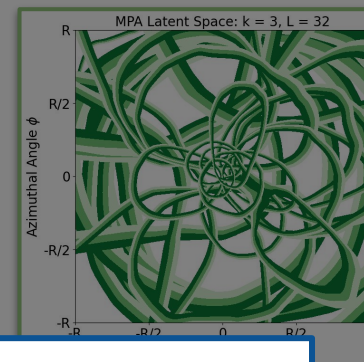
EFN ($k = 1$)
 $L = 128$



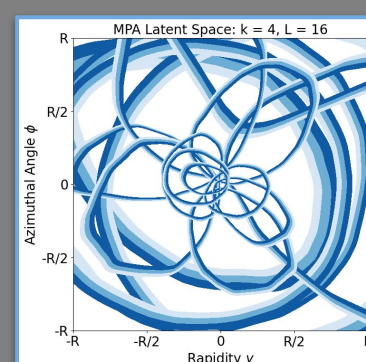
$k = 2$
 $L = 64$



$k = 3$
 $L = 32$



$k = 4$
 $L = 16$



*Interestingly, the performance is constant in the *effective* latent dimensions and number of parameters. Ask me about this!

Latent Spaces

So what? The information is just being shuffled around. Is 16 latent dimensions any more interpretable than 128?

The machine probably learned something, but I didn't!

I'm allowed to control the ML function space!

Let me just choose one that's easy for me to understand: **One dimensional functions**

Because of latent space compression, even one dimension can contain a lot of info!

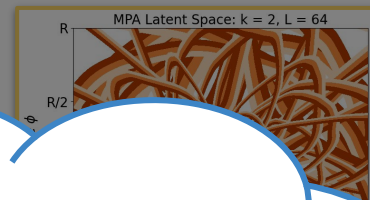
Me, just before the **moment** of clarity.

EFN ($k = 1$)
 $L = 128$

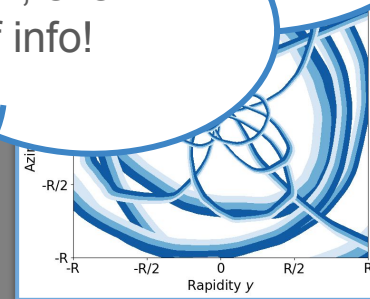
$k = 2$
 $L = 64$

PA Latent Space: $k = 1, L = 128$

MPA Latent Space: $k = 2, L = 64$



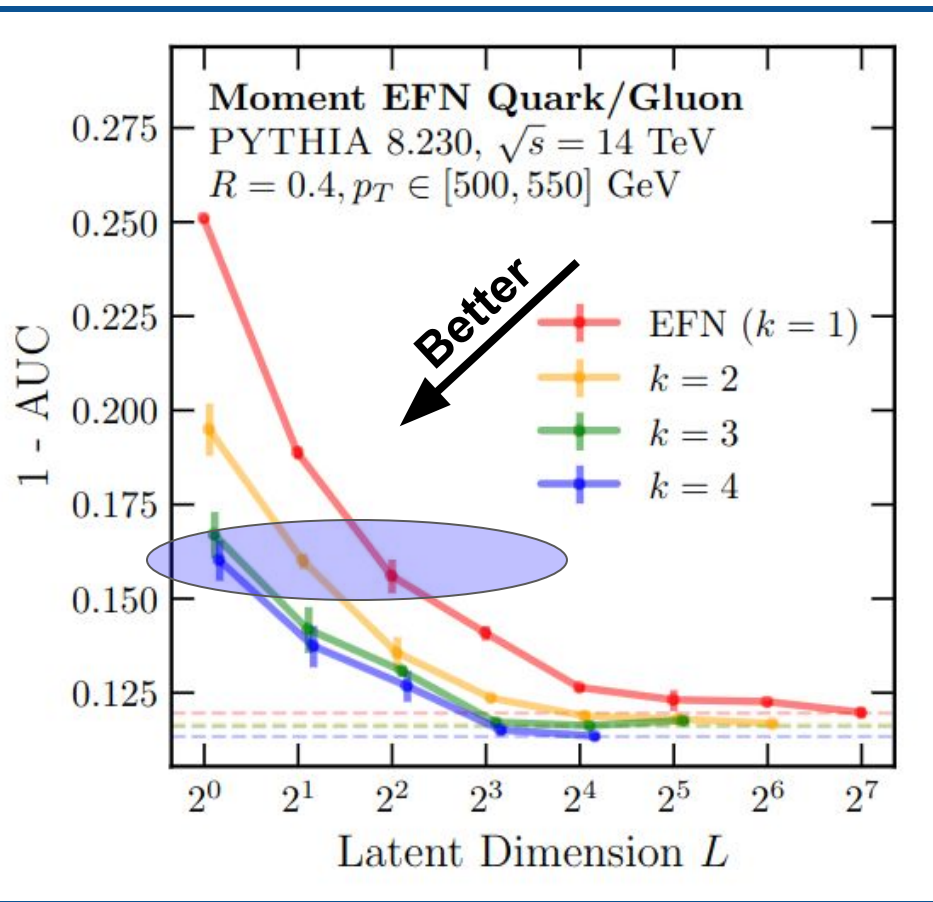
2^3 2^4 2^5
Dimension L



Plots of each of the L different ϕ functions

*Interestingly, the performance is constant in the *effective* latent dimensions and number of parameters. Ask me about this!

Moment Pooling Results (Cont'd)



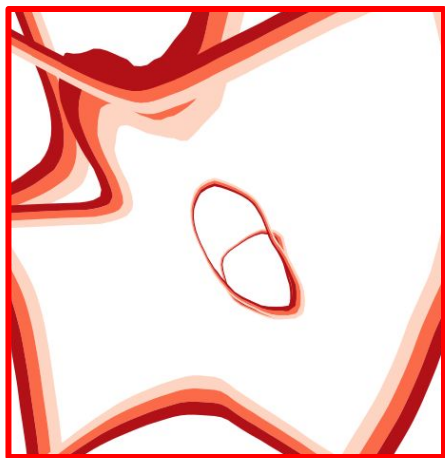
Let's instead look at networks with just a *single* latent dimension.

A **$k = 4$ Moment EFN** with *just one* latent dimension does just as good as an **ordinary EFN** with *four* latent dimensions!

Going down to $L = 1$

The more efficient encoding is extremely useful when we can get to $L = 1$!
Look how much simpler allowing neural networks to multiply makes things!

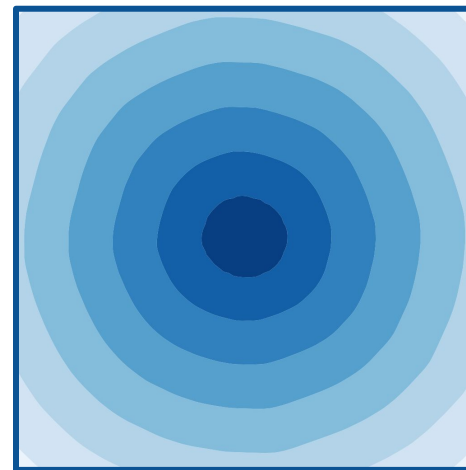
EFN ($k = 1$)
 $L = 4$



Same information!



Order $k = 4$
 $L = 1$



- 4 Different Functions
- Combined in a complicated way
- No symmetry
- ???

- Single function
- Expected to be, and is, radially symmetric
- We can **interpret** this!

The Moment of Truth

We can extract physics from this!

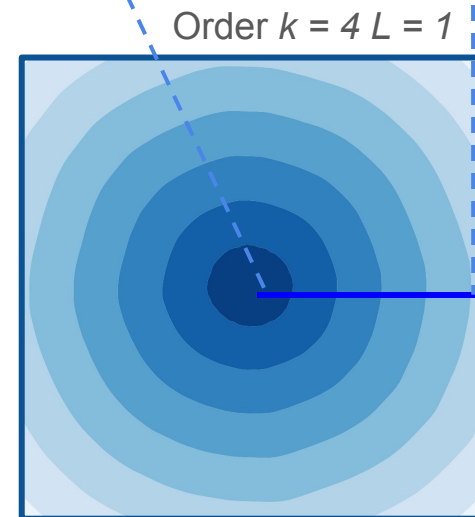
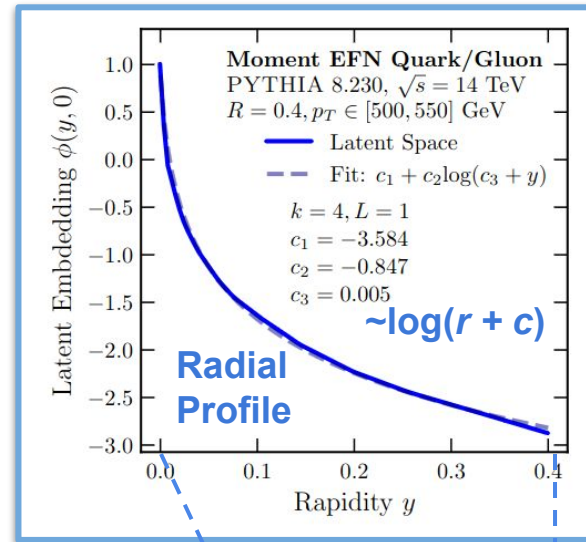
$$\Phi_{\mathcal{L}}(r) = c_1 + c_2 \log(c_3 + r)$$

This defines a **log angularity** observable, related to the $\square \rightarrow 0$ limit of ordinary angularities

We can even see non-perturbative physics!

$$c_3 \sim \frac{\Lambda_{\text{QCD}}}{p_T R} \sim 0.001$$

Just this alone is enough to get an IRC-safe quark-gluon classifier with an AUC ~ 0.83 !



[Aside] Uses in Heavy Ion Jet Quenching

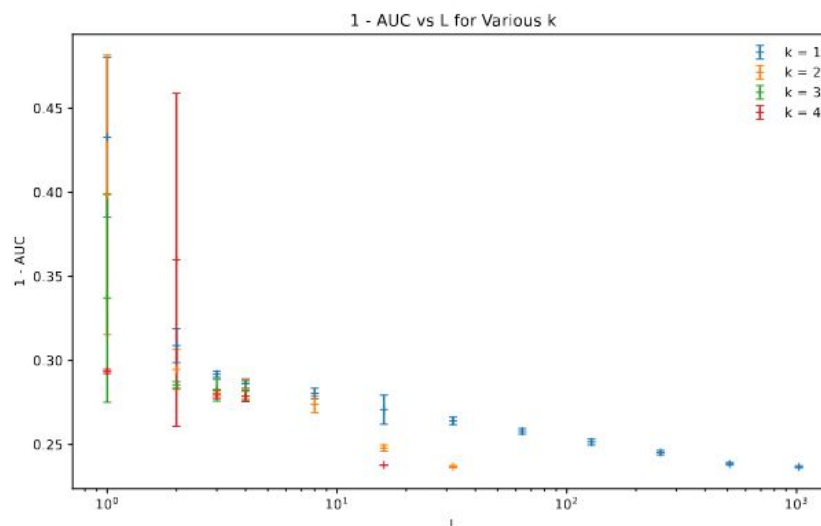
EFNs and Moment EFNs have been used recently to distinguish pp from $Pb-Pb$ collisions to learn the likelihood ratio!

Excited to learn about additional applications of ML in heavy ions from you all!

[[2510.26879](#)]

All is More: Energy Flow Networks for Jet Quenching

João A. Gonçalves



Jet quenching, the modification of jets by the quark-gluon plasma in heavy-ion collisions, provides a sensitive probe of the properties of the medium. A jet-by-jet discrimination study between proton-proton and lead-lead jets using energy flow networks and simple baselines, explicitly retaining medium response and underlying event contamination is presented. As references, linear discriminants and neural networks have been trained on high-level observables such as N -subjettiness and energy flow polynomials, including an extended energy flow polynomial set, in order to quantify the achievable performance without constituent-level learning. Energy flow networks are then trained on jet constituents and extended to observable-enhanced energy flow networks that concatenate standardized N -subjettiness and/or energy flow polynomials to the energy flow network latent space. In the realistic scenario, including both underlying event contamination and medium response, observable-enhanced energy flow networks set state-of-the-art performance with receiver operating characteristic area under the curve of ≈ 0.83 , improving markedly over linear and non-linear baselines and previous work with different architectures. Finally, results from moment energy flow networks, an energy flow network variant that attains comparable area under the curve with a substantially more compact and interpretable latent space are shown. These results establish energy-flow-network-based approaches (especially when enhanced with physics-motivated observables) as practical and robust tools for jet-quenching studies.

~~Jets are complicated~~ can be simple

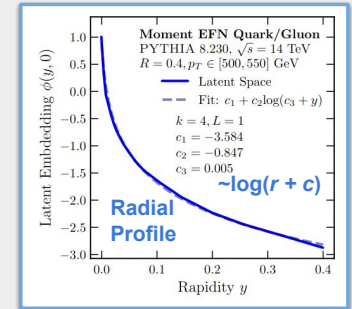
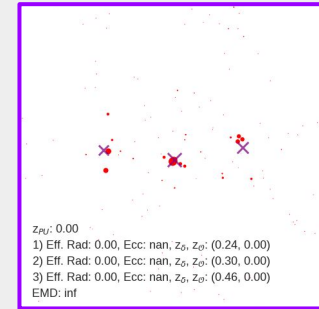
Conclusion

We can design observables, *machine-learned* or *otherwise*, that take advantage of the inherent geometry of jets and QCD in order to make predictions.

These can be crafted to fit a variety of physics goals!

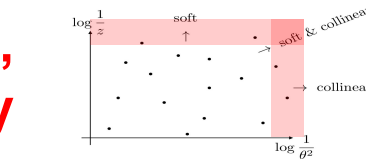
Part 0

Intro: **IRC-Safety, Jets, and Geometry**



Part I

Can you Hear the **Shape** of a Jet?



[Ba, Dogra, **RG**, Tasissa, Thaler, [2302.12266](#)]

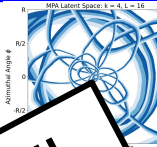
[Ba, Dogra, **RG**, Tasissa, Thaler, [2302.12266](#)]

[**RG**, Thaler, Wu, WIP]

[**RG**, Larkoski, Thaler, [2410.05379](#)]

Part II

Moments of Clarity:
Streamlining Latent Spaces



[**RG**, Osathapan, Thaler, [2403.08854](#)]

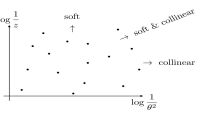
Part III
[You are Here!]

Conclusion

Email me questions at gambhirb@ucmail.uc.edu!

- [Ba, Dogra, **RG**, Tasissa, Thaler; [2302.12266](#)]
- [**RG**, Thaler, Wu; WIP]
- [**RG**, Larkoski, Thaler; [2410.05379](#)]
- [**RG**, Osathapan, Thaler; [2403.08854](#)]
- [Du Plessis, **RG**; WIP]
- [**RG**, LeBlanc, Zhou; [2509.19431](#)]
- [**RG**, Larkoski; WIP]

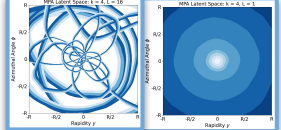
IRC-Safety, Jets, and Geometry



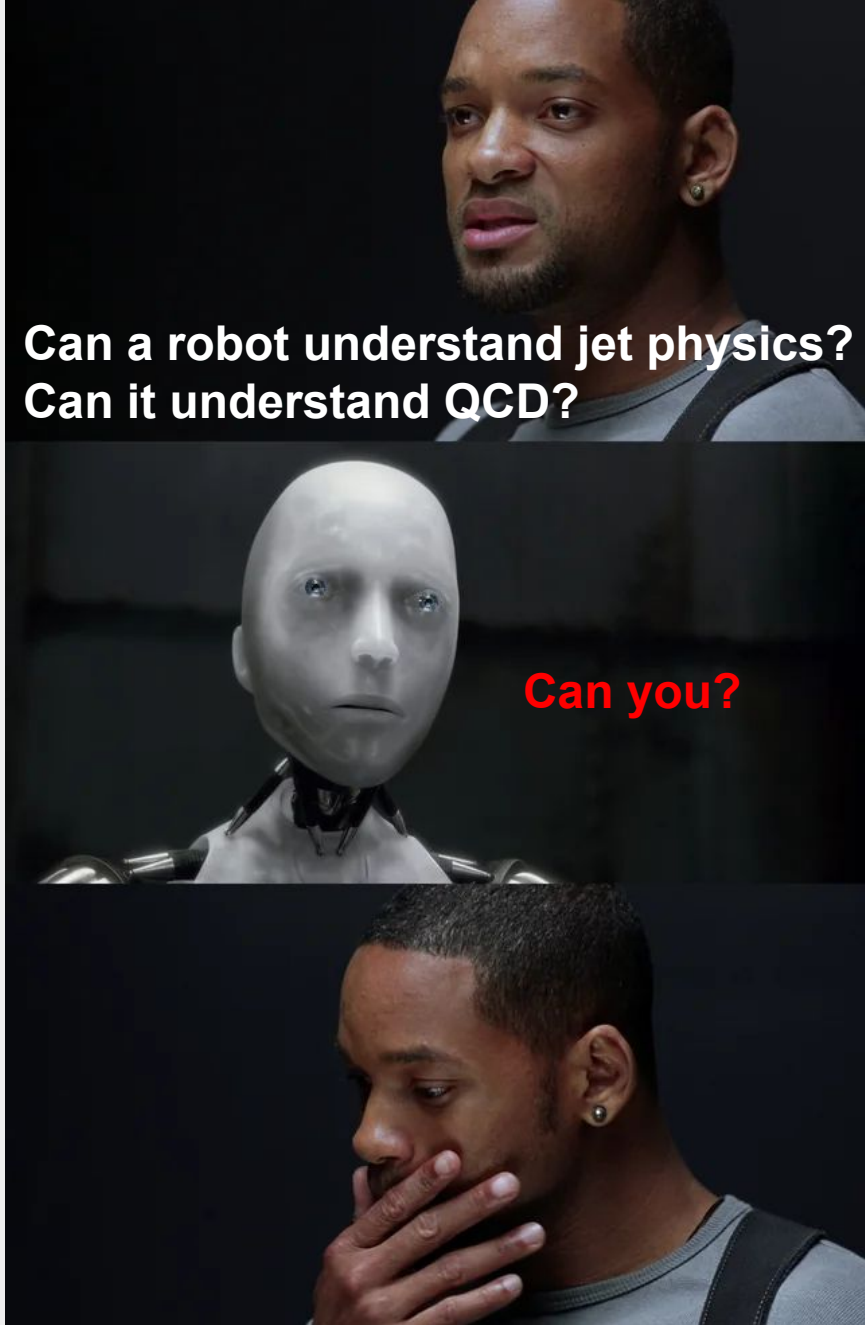
Can you Hear the Shape of a Jet?



The Moments of Clarity:
Streamlining Latent Spaces



Any Questions?



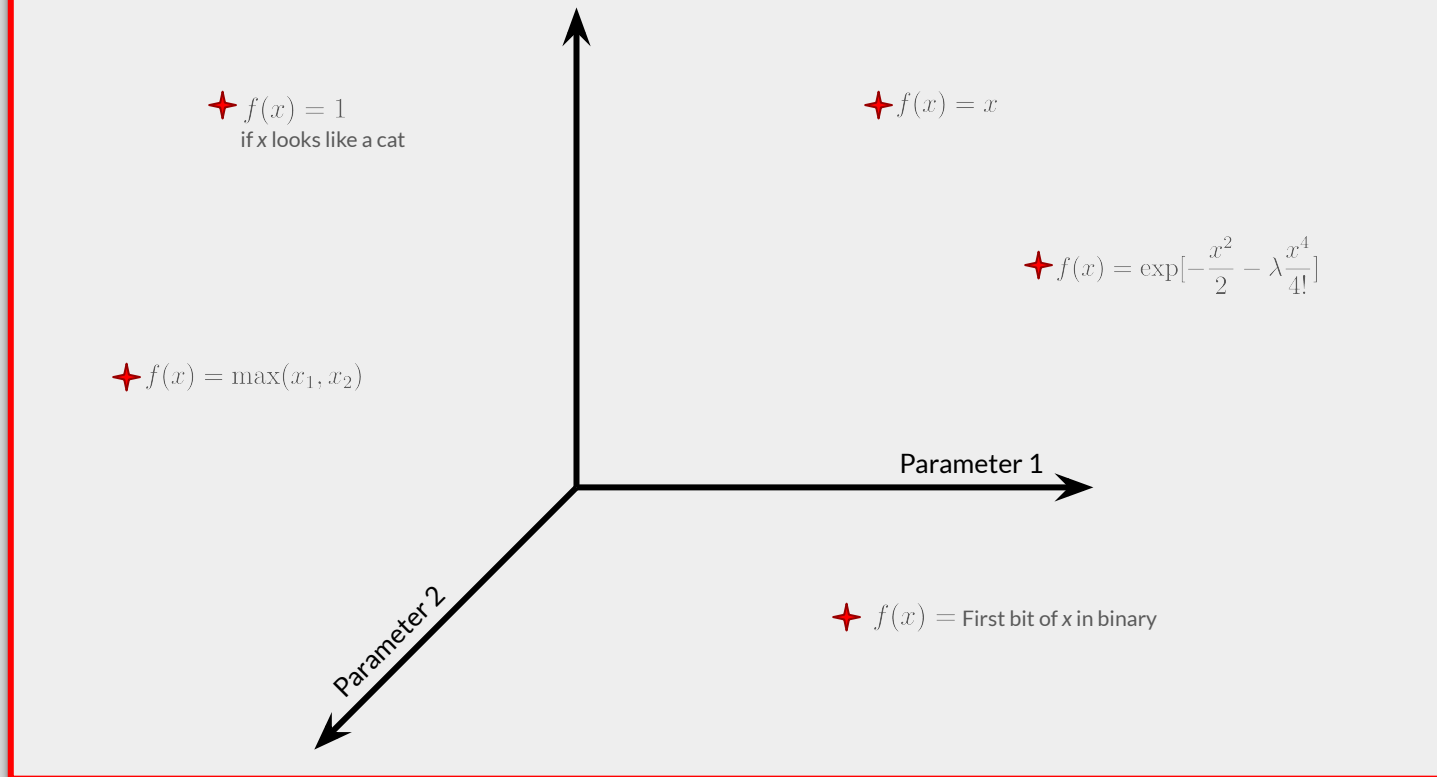
Can a robot understand jet physics?
Can it understand QCD?

Can you?

Backup

What even is Machine Learning (Rikab's Take)?

Ingredient 1: Function Space



Machine Learning (ML) = A method to *numerically* explore the space of functions

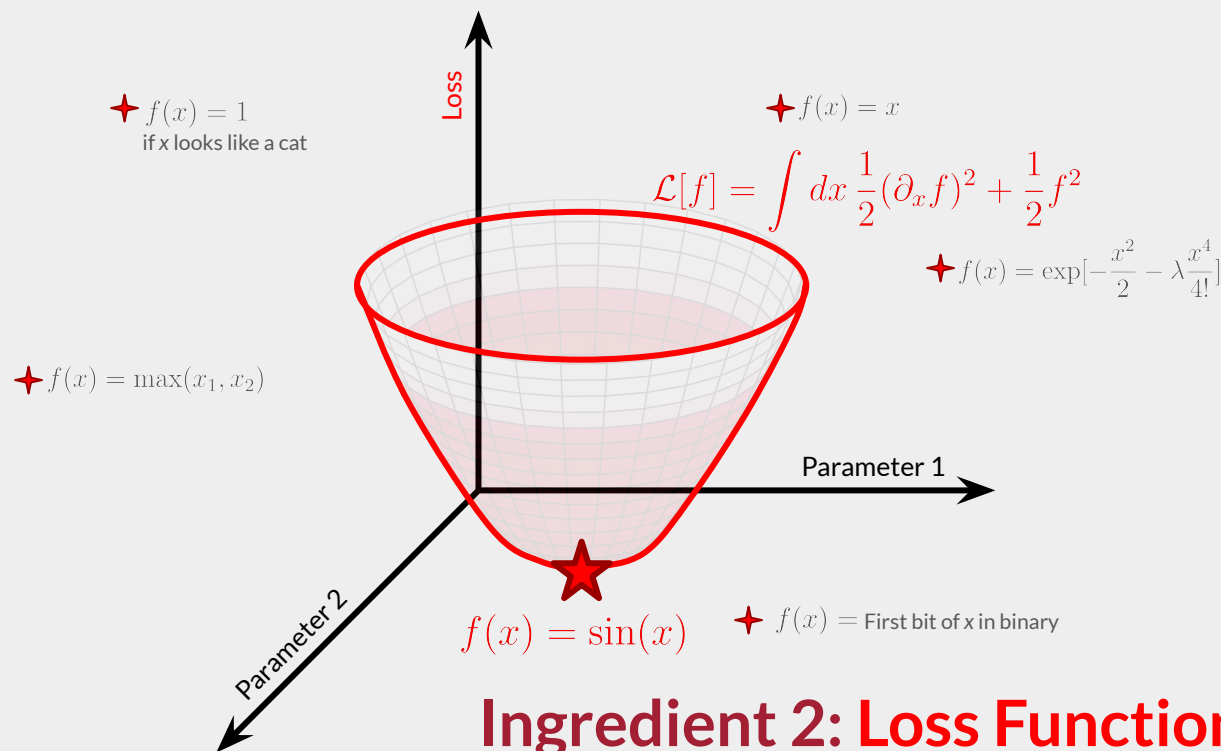
Key workhorse: **Universal Approximation Theorem(s)**

Large classes of functions can be approximated by *Neural Networks* = Parametrized Functions

For this talk, the details of how this approximation is constructed (layers, activations, ...) don't matter!

What even is Machine Learning (Rikab's Take)?

Ingredient 1: Function Space



Ingredient 2: Loss Function(al)

Many problems in life (physics and statistics) involve taking **minimums** over function space!

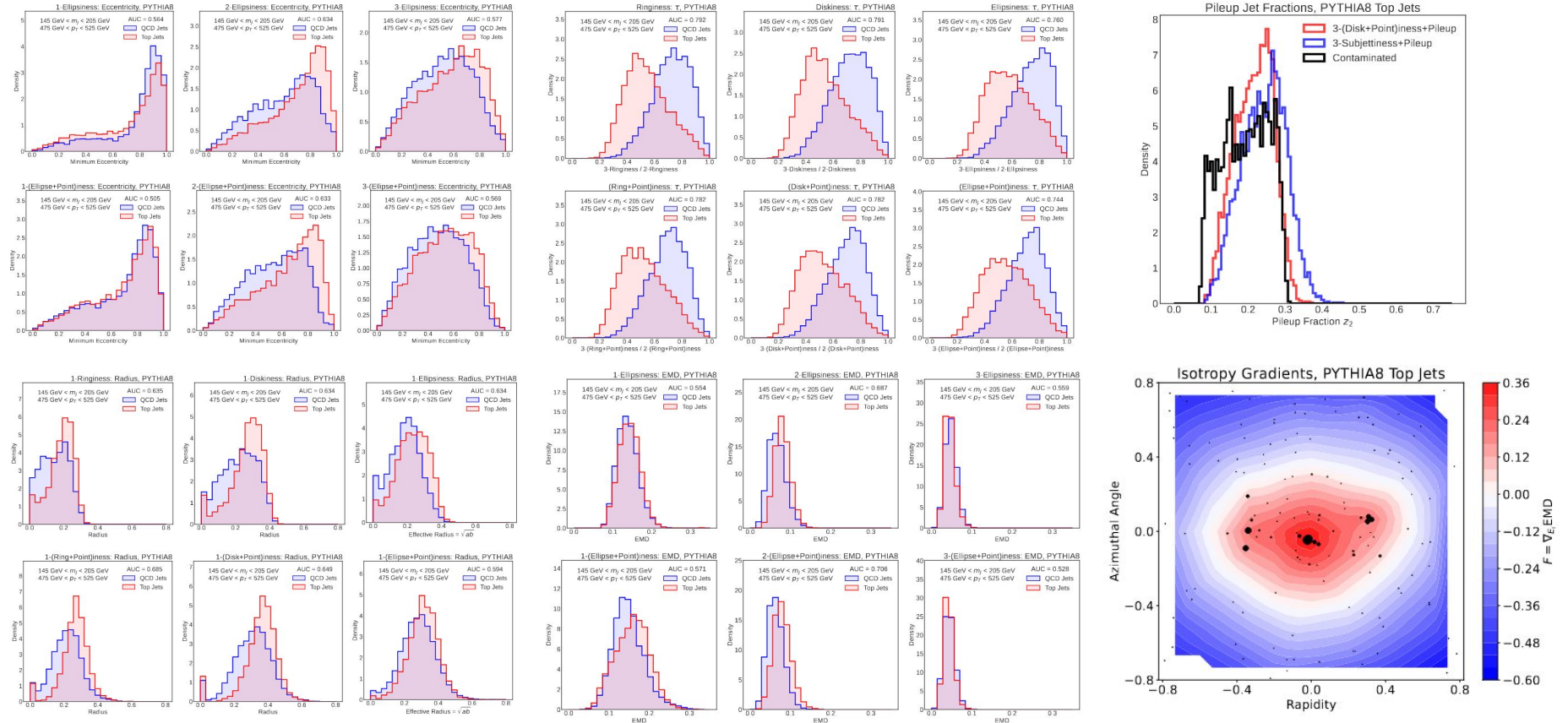
This is *identical* to how we do Lagrangian Mechanics, down to using the Euler-Lagrange equations.

Statistical Inference (Part I), Defining Observables (Part II), Classification (Part III), even Resummation (Part IV)

ML lets us do functional minimization *numerically*!

The details of how we do this (gradient descent, autodifferentiation, ...) are not important for this talk.

New IRC-Safe Observables



... **Lots** of extractable information!

Automatic Grooming with Shapes

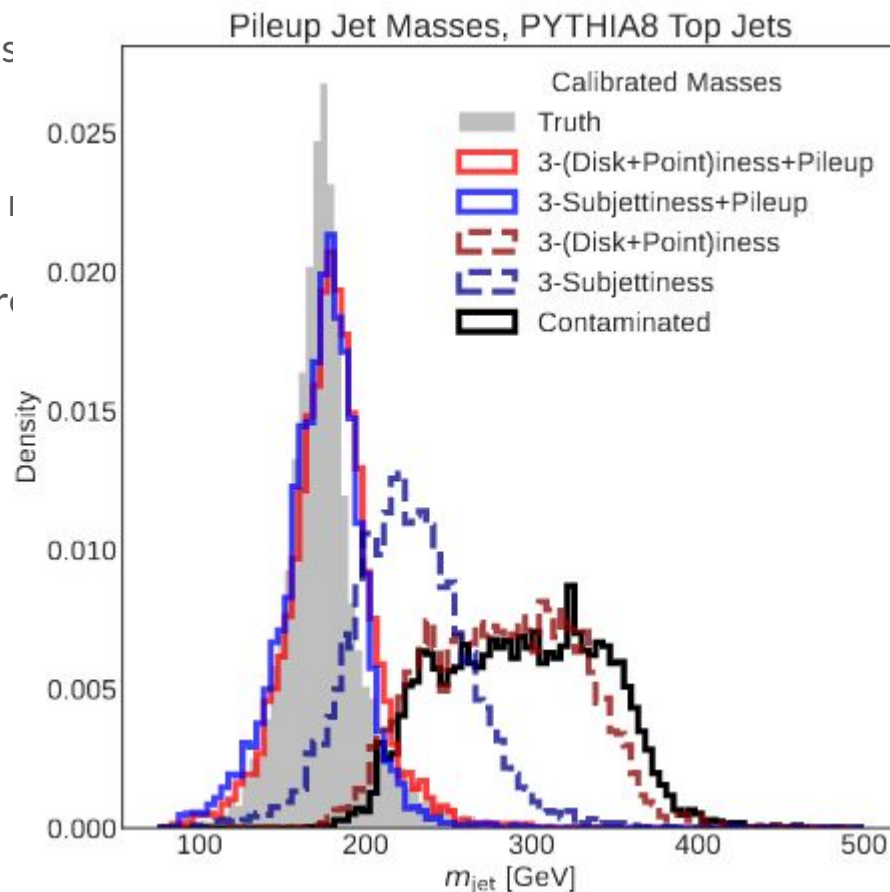
Use shapes to approximate events and extract mass with floating weight!

No external hyperparameters, unlike softdrop. Only 1

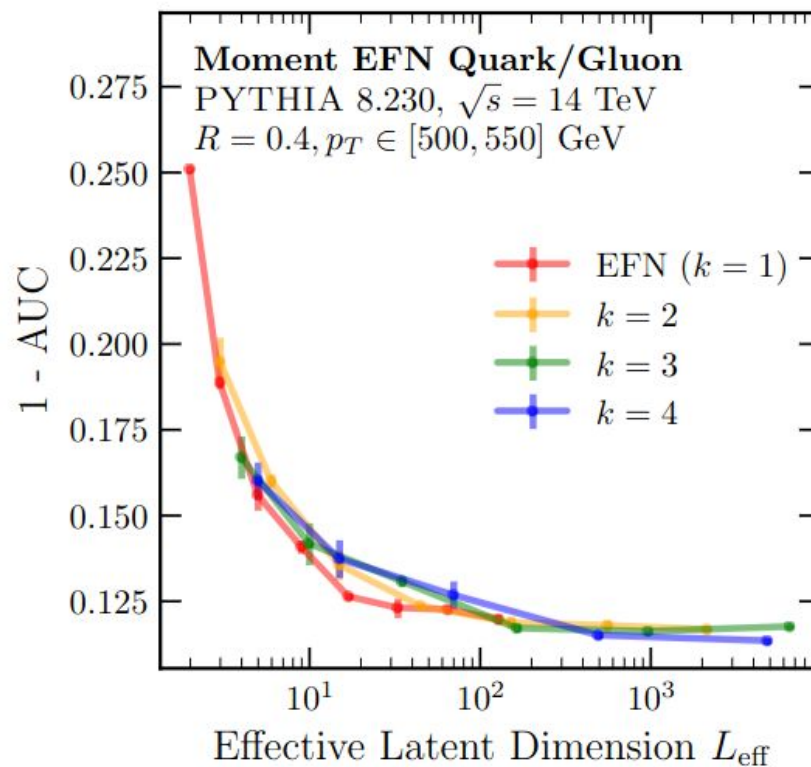
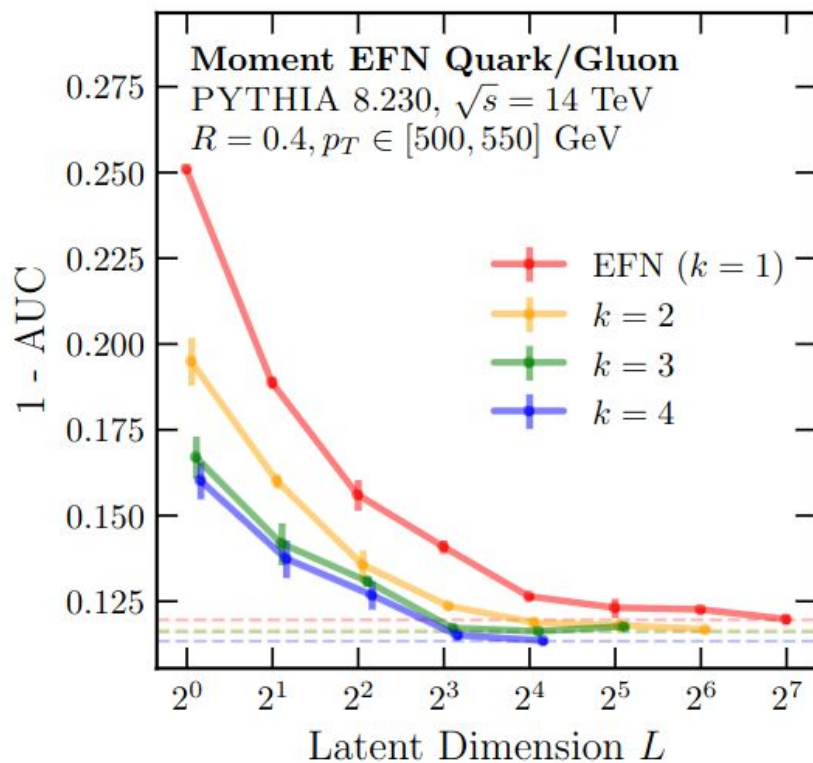
Contaminate top jets with 5-30% extra energy spr

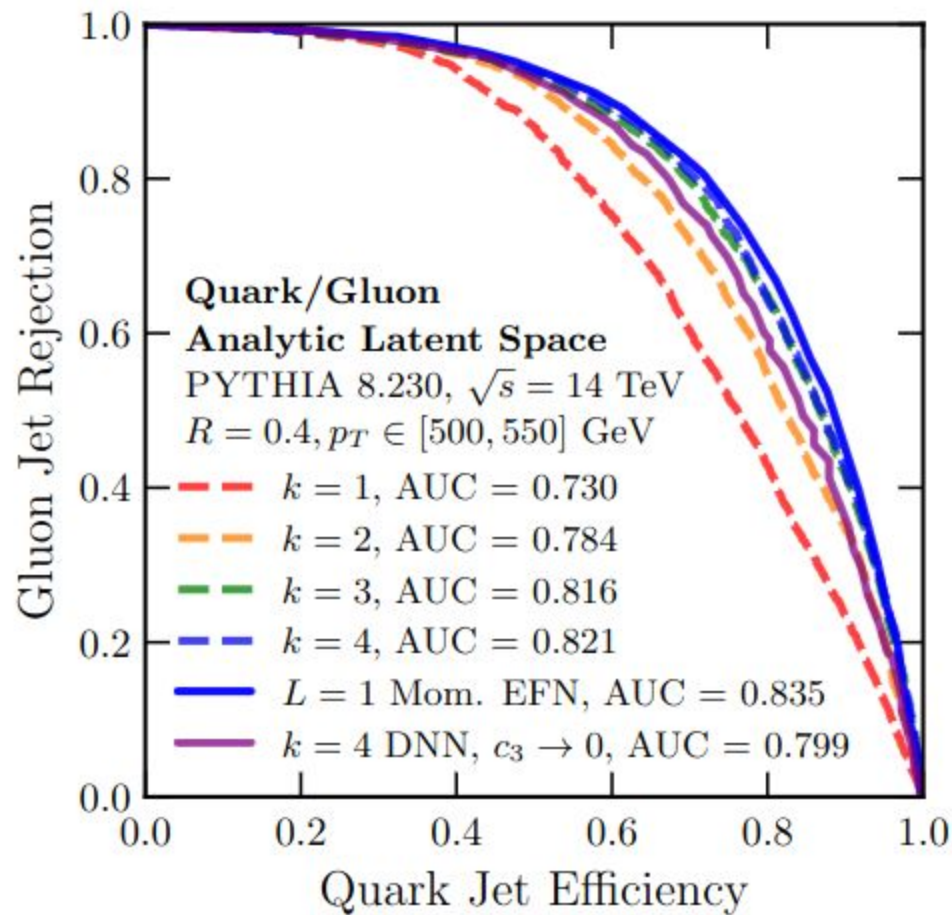
Consider 4 shapes:

- 3-Subjettiness
- 3-Subjettiness + Pileup
- 3-(Disk+Point)iness
- 3-(Disk+Point)iness + Pileup

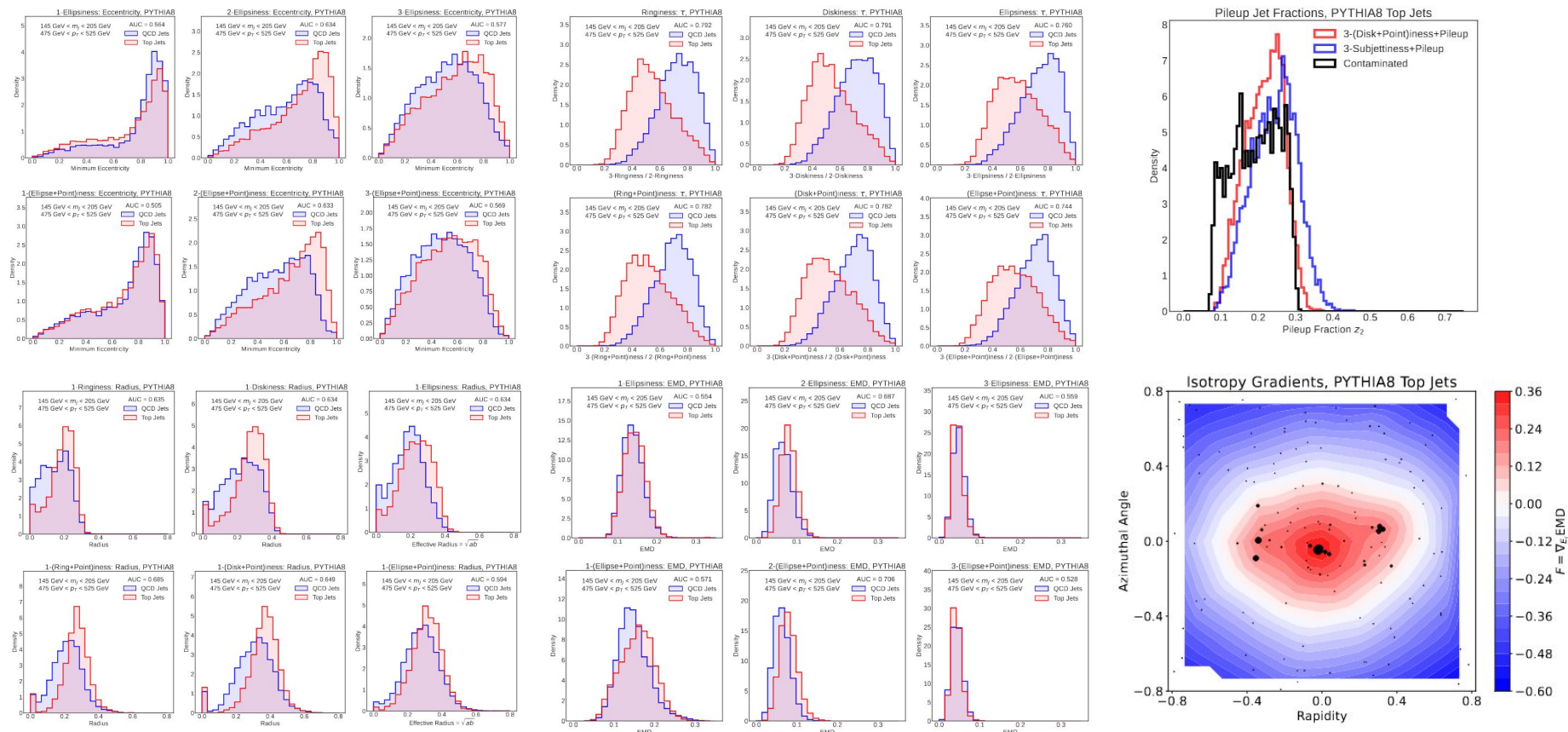


Can also consider ellipses instead of disks – only marginally better performance





New IRC-Safe Observables



... **Lots** of extractable information!

Automatic Grooming with Shapes

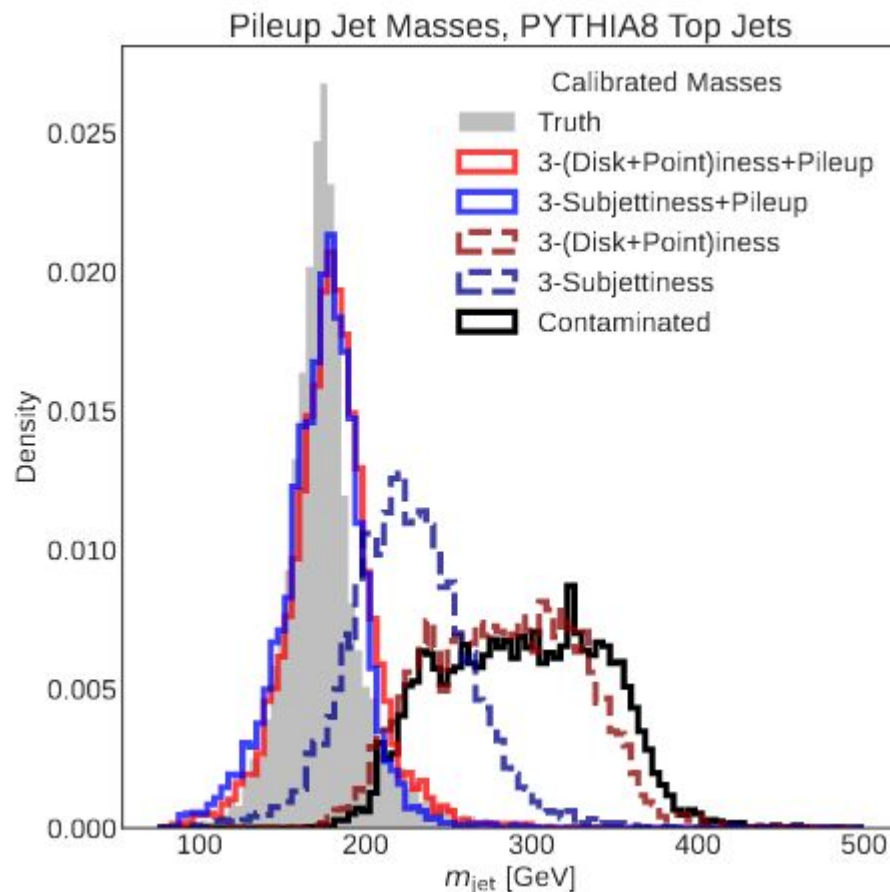
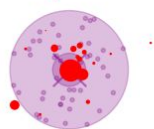
Use shapes to approximate events and extract masses – model pileup with a uniform background with floating weight!

No external hyperparameters, unlike softdrop.
Only need to assume pileup is uniform!

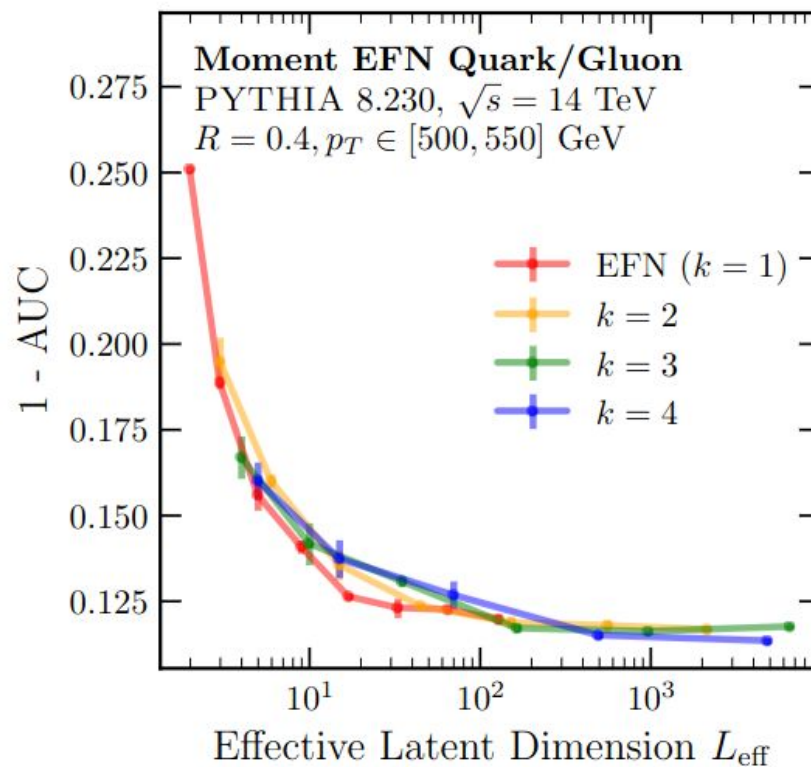
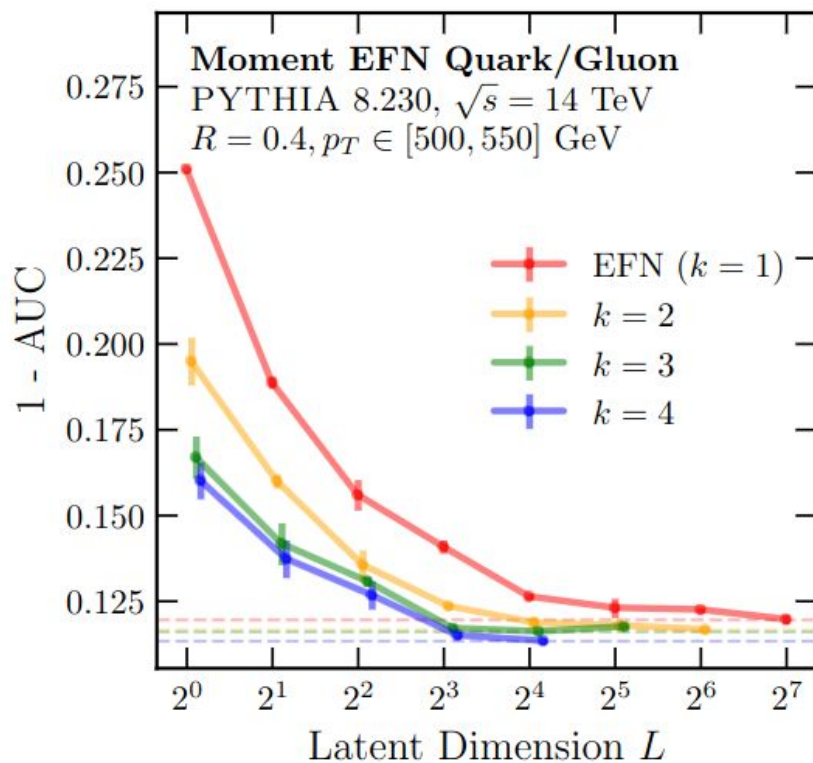
Contaminate top jets with 5-30% extra energy spread uniformly in an 0.8×0.8 plane

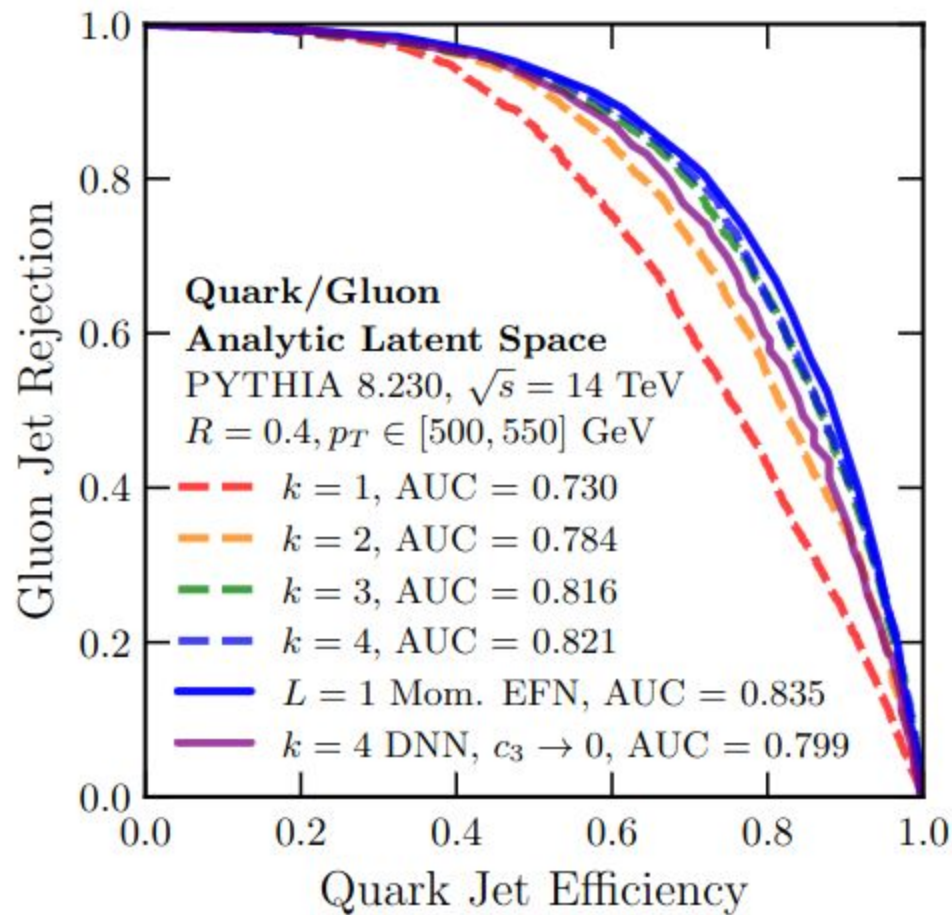
Consider 4 shapes:

- 3-Subjettiness
- 3-Subjettiness + Pileup
- 3-(Disk+Point)iness ←
- 3-(Disk+Point)iness + Pileup



Can also consider ellipses instead of disks – only marginally better performance





The Spectral EMD

Similar to the ordinary EMD, the **Spectral EMD (SEMD)**¹ is an IRC-saf metric between events or jets, computed on their **spectral representations**:

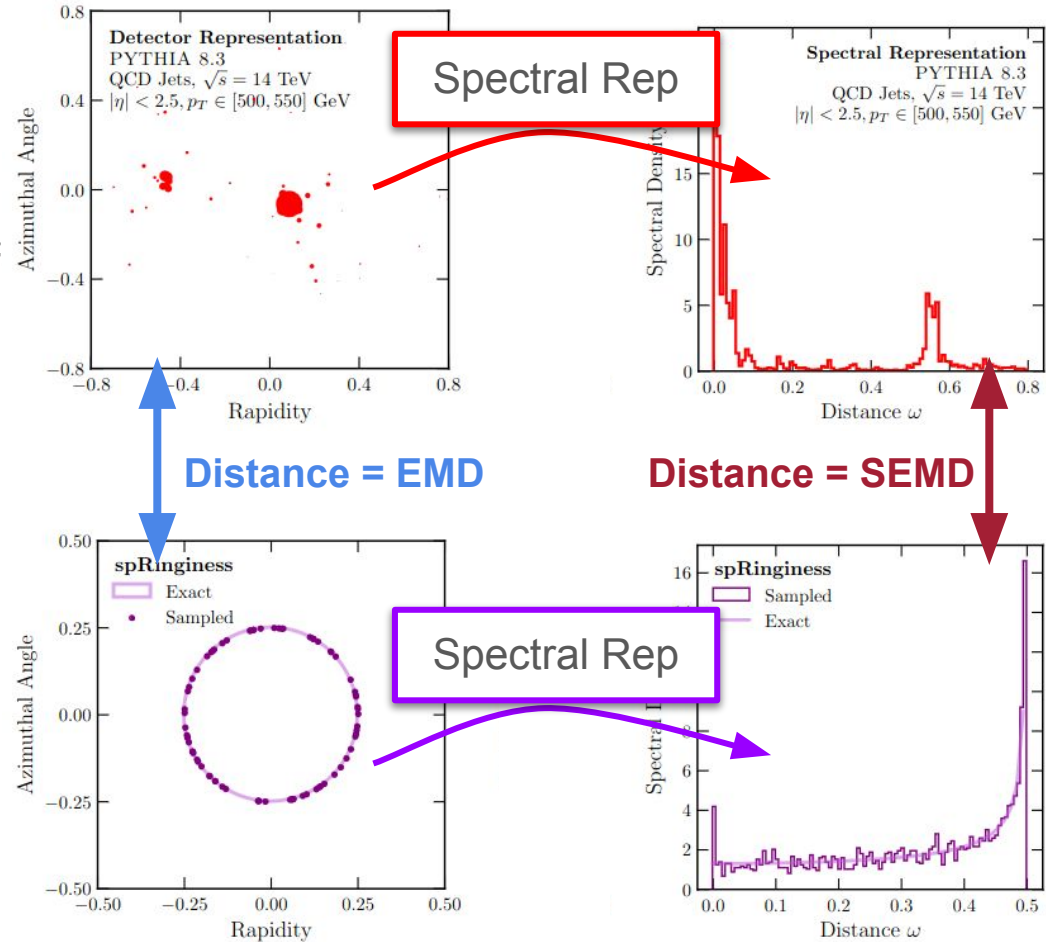
$$s(\omega) \equiv \sum_{i,j \in \mathcal{E}} E_i E_j \delta(\omega - \omega_{ij})$$

= list of energy-weighted pairwise distances

Then the p -SEMD is given by:

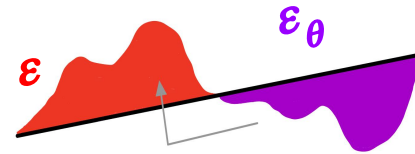
$$\text{SEMD}_p(s_A, s_B) \equiv \int_0^{E_{\text{tot}}^2} dE^2 |S_A^{-1}(E^2) - S_B^{-1}(E^2)|^p$$

Inverse of integral of $s(\omega)$



The SEMD automatically respects all isometries of the metric ω . In this case, the SEMD is invariant under rotations or translations of either event or jet

Technical Details ...



For $p = 2$, possible problem on the sp

For events A, B , the p **spectral EMD** is defined as (1D OT!):

EMD = Work done to move "dirt" optimally

$$\text{SEMD}_{\beta,p}(s_A, s_B) \equiv \int_0^{E_{\text{tot}}^2} dE^2 |S_A^{-1}(E^2) - S_B^{-1}(E^2)|^p$$

$$\begin{aligned} \text{SEMD}_{\beta,p=2}(s_A, s_B) = & \sum_{i < j \in \mathcal{E}_A} 2E_i E_j \omega_{ij}^2 + \sum_{i < j \in \mathcal{E}_B} 2E_i E_j \omega_{ij}^2 \\ & - 2 \sum_{n \in \mathcal{E}_A^2, l \in \mathcal{E}_B^2} \omega_n \omega_l (\min[S_A(\omega_n^+), S_B(\omega_l^+)] - \max[S_A(\omega_n^-), S_B(\omega_l^-)]) \\ & \times \Theta(S_A(\omega_n^+) - S_B(\omega_l^-)) \Theta(S_B(\omega_l^+) - S_A(\omega_n^-)), \end{aligned}$$

S = **cumulative spectral function**

$\pm$ indicates whether or not to include ω in the sum

The trick: Sum over pairs n of particles within each event.

Looks like $O(N^4)$, but with clever sorting & indexing in 1D, reduces to **$O(N^2)$** !

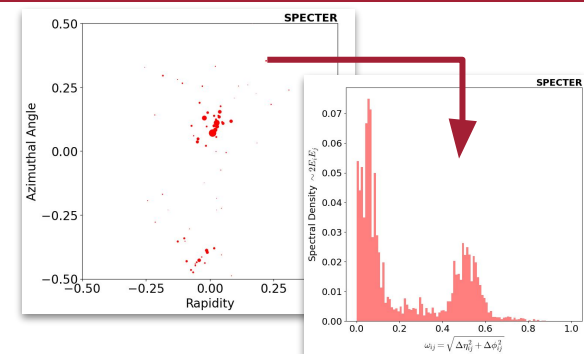
The **spectral density function**

$$s(\omega) = \sum_{i=1}^N \sum_{j=1}^N E_i E_j \delta(\omega - \omega(\hat{n}_i, \hat{n}_j))$$

Pairwise Distances

Reduces events to 1D, while preserving all* information about the event, up to translations and rotations.

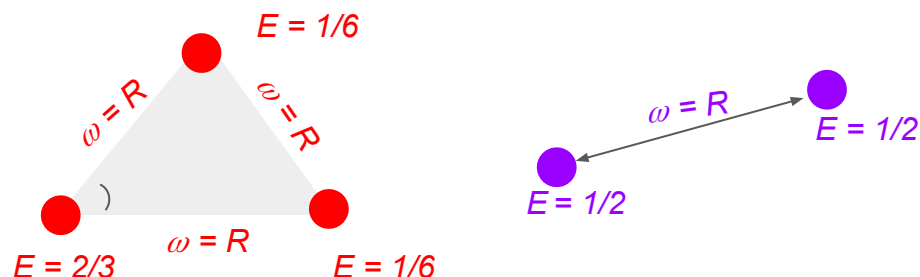
*up to measure 0, but important degeneracies, ask me about this later!



EMDs, ...

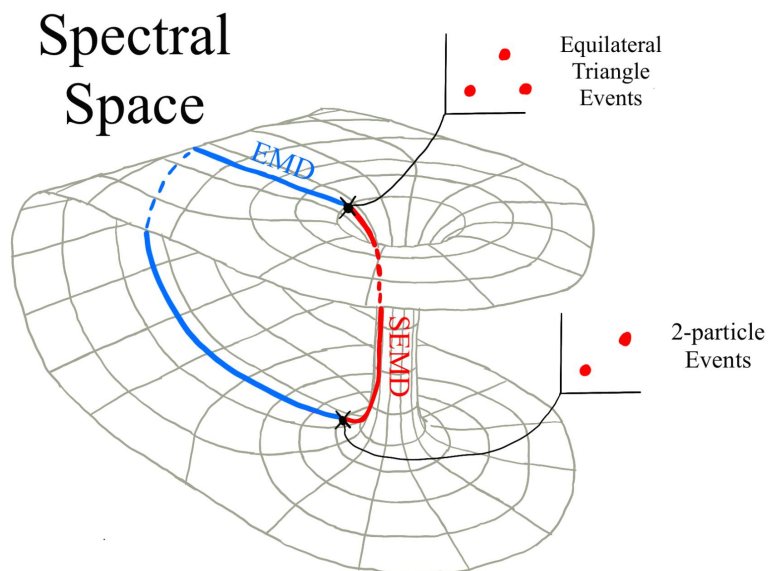
The EMD vs. The SEMD

The SEMD is *topologically different* from the ordinary EMD!



This three particle event looks very different from this two particle event, and thus they have a large EMD.

But their SEMD is zero! The spectral function only cares about pairwise distances and degenerate configurations can occur.



The space of events gets “pinched” at degenerate configurations when looking at only their spectral representations

This can come into play when events have 3 hard prongs, e.g. top jets!

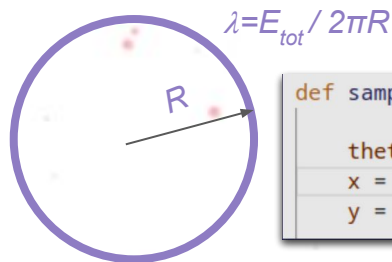
Full Example: How “ring-like” are jets?

SPECTER

Our code framework
for these calculations



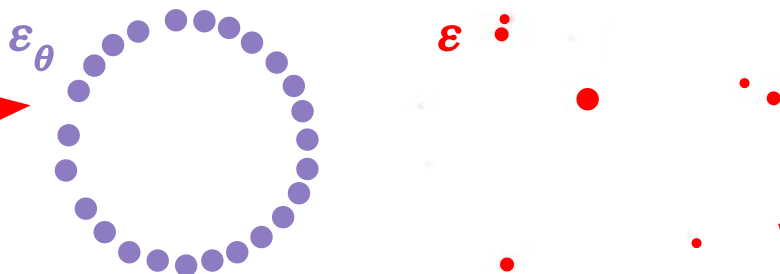
Step 1: Define the shape with parameters



```
def sample_circle(params, N, seed):  
    thetas = jax.random.uniform(seed, shape=(N,))  
    x = params["Radius"] * jnp.cos(thetas)  
    y = params["Radius"] * jnp.sin(thetas)
```

Unlike ordinary EMD, not necessary to specify center / orientation!

Step 2: Sample from Parameterized Shapes



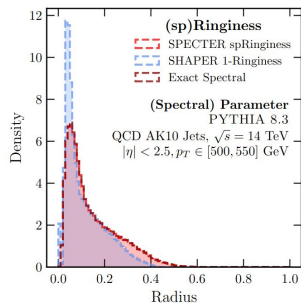
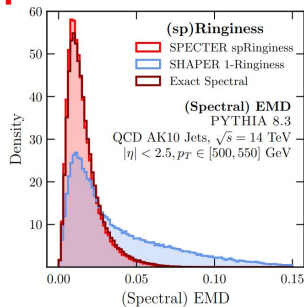
Step 3: Calculate the spectral metric between events and shapes

$$\text{SEMD}_{\beta,p=2}(s_A, s_B) = \sum_{i < j \in \mathcal{E}_A} 2E_i E_j \omega_{ij}^2 + \sum_{i < j \in \mathcal{E}_B} 2E_i E_j \omega_{ij}^2 - 2 \sum_{n \in \mathcal{E}_A^2, l \in \mathcal{E}_B^2} \omega_n \omega_l (\min[S_A(\omega_n^+), S_B(\omega_l^+)] - \max[S_A(\omega_n^-), S_B(\omega_l^-)]) \times \Theta(S_A(\omega_n^+) - S_B(\omega_l^-)) \Theta(S_B(\omega_l^+) - S_A(\omega_n^-)) ,$$

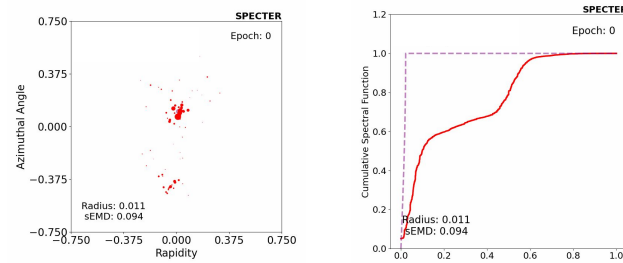
Key difference from previous work: We use the SEMD, *not* the EMD!

Pictured: 100k Jets, PYTHIA 8 QCD Jets

Step 5: Plots!



Step 4: Minimize w.r.t. parameters using grads



Pictured: Animation of optimizing for the radius R

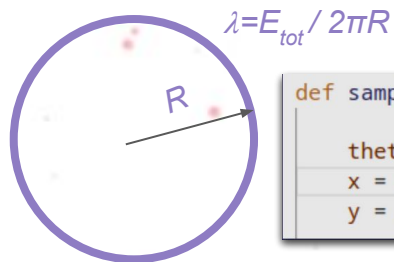
Full Example: How “ring-like” are jets?

SPECTER

Our code framework
for these calculations



Step 1: Define the shape with parameters



Unlike ordinary EMD, not necessary to s

```
def samp
    theta
    x = p
    y = p
```

The spectral EMD, and its optimization, are often partially or **completely solvable** in closed form!

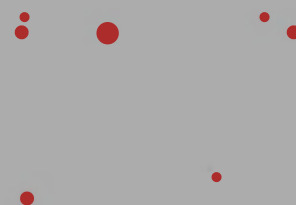
$$R_{\text{opt}} = \frac{2}{\pi} \sum_{\substack{n \in \mathcal{E}^2 \\ \omega_n < \omega_{n+1}}} \omega_n \left[\sin \left(\frac{\pi}{2E_{\text{tot}}^2} \sum_{\substack{n \leq m \in \mathcal{E}^2 \\ \omega_m < \omega_{m+1}}} (2EE)_m \right) - \sin \left(\frac{\pi}{2E_{\text{tot}}^2} \sum_{\substack{n < m \in \mathcal{E}^2 \\ \omega_m < \omega_{m+1}}} (2EE)_m \right) \right]$$

$$\text{SEMD}_{\beta, p=2}(s, s_{\text{jet ring}}) = \sum_{i < j \in \mathcal{E}} 2E_i E_j \omega_{ij}^2 - 2E_{\text{tot}}^2 R_{\text{opt}}^2$$

For many shapes, we can completely short circuit having to perform expensive optimization over an optimal transport problem entirely!

Alternatively...

Step 2: Sample from Parameterized Shapes



the spectral metric between

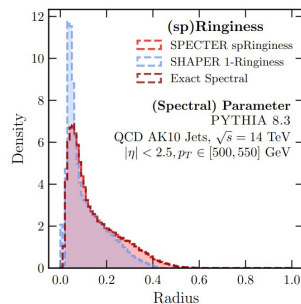
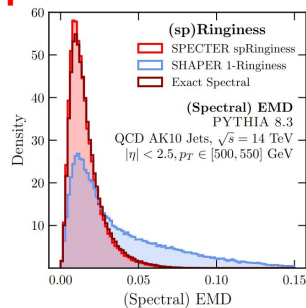
$$2E_i E_j \omega_{ij}^2 + \sum_{i < j \in \mathcal{E}_B} 2E_i E_j \omega_{ij}^2$$

$$(\omega_n^+, S_B(\omega_l^+)) - \max [S_A(\omega_n^-), S_B(\omega_l^-)]$$

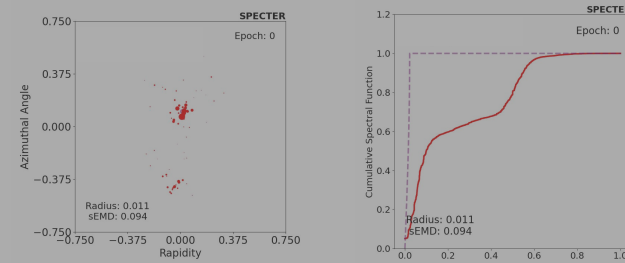
$$- S_B(\omega_l^-)) \Theta (S_B(\omega_l^+) - S_A(\omega_n^-)) ,$$

Pictured: 100k Jets, PYTHIA 8 QCD Jets

Step 5: Plots!



Step 4: Minimize w.r.t. parameters using grads



Pictured: Animation of optimizing for the radius R

Hearing Jets (sp)Ring

Runtimes (NVIDIA A100 GPU):

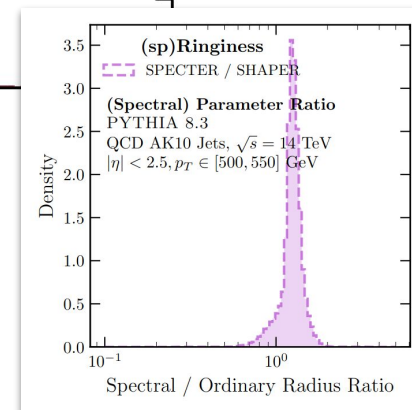
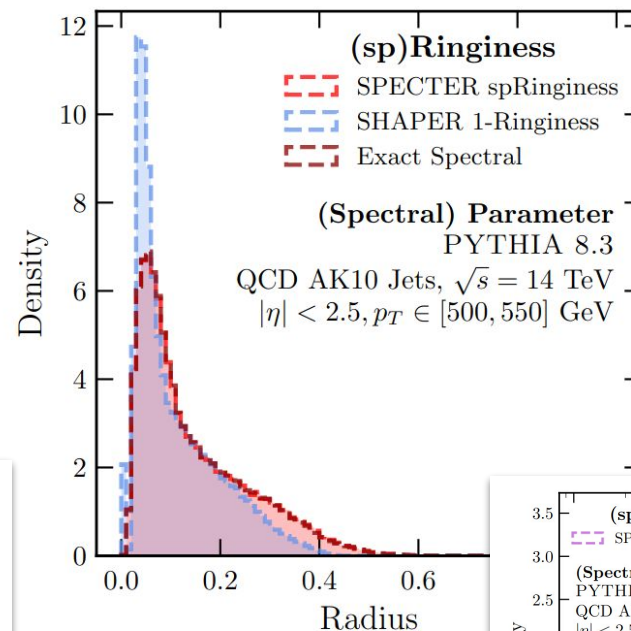
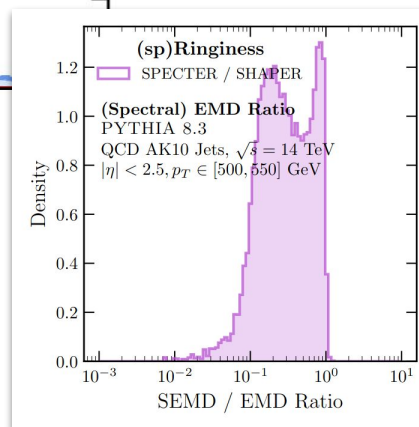
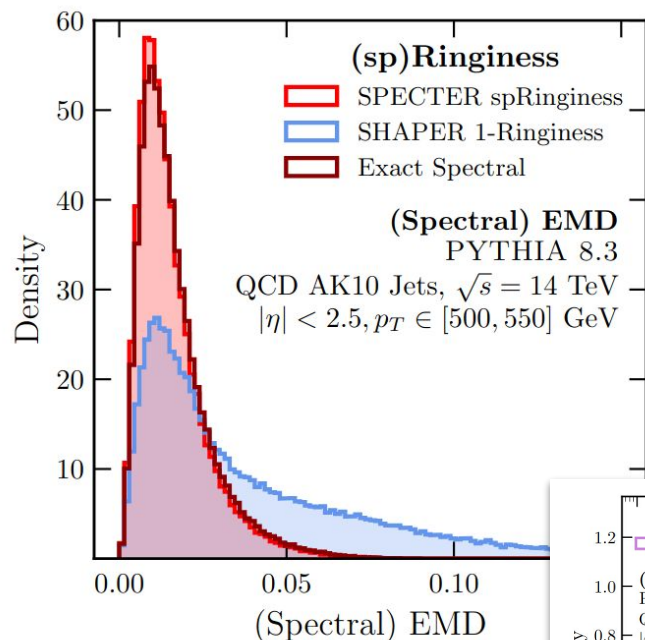
SHAPER (EMD): ~ 36 hours / 100k events

Generalized SPECTER: ~55 seconds / 100k events

Closed Form SPECTER: ~ < 0.1 seconds / 100k events

The SEMD and EMD are qualitatively different, but give similar radii!

They probe the same event length scale

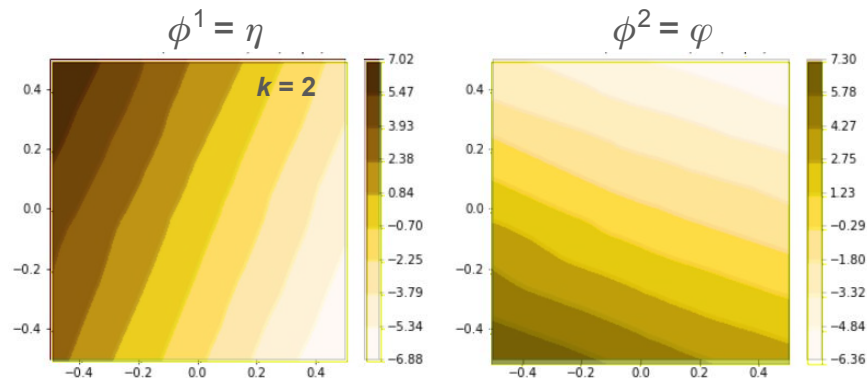


Multiplication is all You Need

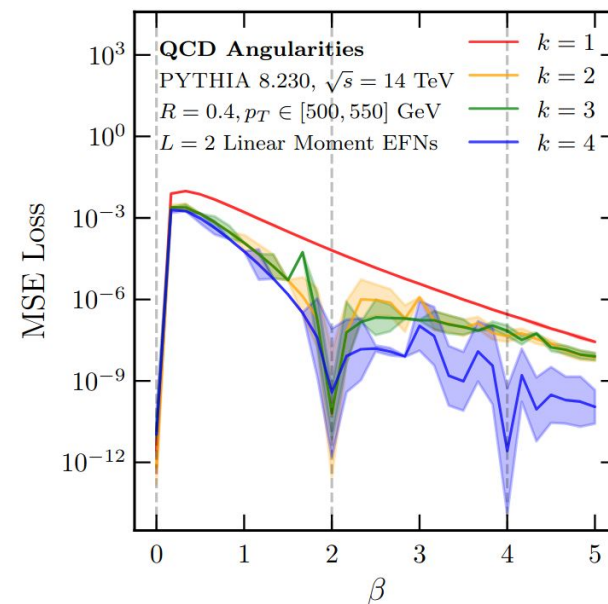
In the moment language, even integer^{*} β jet angularities $\Leftrightarrow k = \beta^{\text{th}}$ moments!

$$\begin{aligned}\lambda^{(\beta)}(\mathcal{P}) &= \sum_i z_i (\eta_i^2 + \phi_i^2)^{\beta/2} \\ &= \langle \eta^\beta \rangle_{\mathcal{P}} + \langle \phi^\beta \rangle_{\mathcal{P}} + \text{Cross Moments}\end{aligned}$$

Learns the simplest latent representations!



Encourage learning a diagonal basis in (η, ϕ) using L1 Regularization on F



The lesson: Basic observables are naturally represented with multiplication, so build it in!